

BEYOND PEN AND PAPER: MODERNIZIING THE LABORATORY MANAGEMENT OF
THE COLLEGE OF ENGINEERING AT BENEDICTO COLLEGE

A System Study presented to the
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of the Requirement for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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ABSTRACT

This research aimed to develop and evaluate a modernized laboratory management system for the College of Engineering at Benedicto's College in order to modernize and add a new way in performing the different transactions within the engineering laboratories such as their inventory monitoring, borrow and return transactions as well as their laboratory violation and penalties transactions. The system aimed to address the inefficiencies of the current paper-based system by providing a well-structured platform for managing laboratory records, improving security and privacy, and implementing a built-in search functionality in order to easily find and locate the specific record that needed to be checked upon by the user, This provides an advantage to lessen the manual searching and enhance the overall performance as to how the laboratory personnels can do their tasks within the engineering laboratory. A rapid application development (RAD) approach was used, including system prototyping and interview techniques, to gather user feedback and refine the system design. The developed system successfully addressed the research objectives, providing a more efficient and secure solution for laboratory management. The findings have practical implications for improving laboratory operations and enhancing the overall performance of the engineering department.

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CHAPTER I

Introduction

A. Rationale of the Study

The College of Engineering's current laboratory management system is outdated and inefficient, which can be prone to confusion. The laboratory records play a vital role within the department itself as it holds the information of their apparatus requirements, borrowing and returning of the apparatuses, and the penalty records as well. The employees have tried using other external factors such as Excel, but it has complex functions and limited accessibility because it is an expensive subscription-based software.

The current laboratory management system is ineffective as it is mainly based on the usage of the traditional records logbook for its processes and record-keeping. There are also concerns with the security of information because it is prone to data loss and damages caused by foreign damage factors, the search method is also time-consuming because they have to manually look through and check each page to search for a specific record or information, the appliance of old paperwork procedure during the borrowing and returning process as well as the need of improving the penalty records. If these are left unattended, this can lead to frustration, security concerns as well as the risk of errors and inconsistencies which can affect the performance of the college department itself. The researchers aim to fix these issues by developing a new laboratory management system that is easy to handle and manage.

The new system aims to re-define the outdated laboratory management of the College of Engineering such as their mini-inventory, borrowing-and-returning, as well their penalty records. As of now, everything in the laboratory's management is done mainly on paper-based aspects which can lead to handling inefficiencies. The new system will feature a modernized way of managing the different data records of the Engineering Department such as

their mini-inventory, borrowing-and-returning of apparatuses as well as their penalty records. The assigned working personnel will have better efficiency when managing the laboratory records and processing. The students will also be aware of and can have a better understanding of how the laboratory records can now be managed in a modernized way. Finally, the system aims to boost the security, productivity, and satisfaction of the Engineering Department.

The new system will bring features that can store information on the following laboratory apparatuses and even has a function where they can update the following apparatuses' status or condition. It can also store and handle the information records of the borrowers and when they return the corresponding item both current and in the past as well as it can also store the penalty records of violators and it also has a search functionality that can furthermore help in navigating the different information records.

To fully implement these features, the researchers and developers will need to take a reasonable amount of time which is estimated at around eight to nine months of development in order to make the system functional and ready. The development and study will only be conducted within Benedicto College, Mandaue Campus, specifically within the College of Engineering Department. The research environment will not include the other branches of Benedicto College.

B. Objectives of the System

The researchers aim to develop and evaluate a new and modernized Laboratory Management system that will address the inefficiencies and concerns of the current laboratory management process of the College of Engineering. This system aims to:

- create a well-structured platform that can let the assigned personnel manage the different laboratory records such as its inventory, borrow-and-returning information as well as penalty records;

- minimize the traditional paper-based work and improve its processing to achieve a technological and modernized way of record-keeping management;
- improve security and privacy by implementing relevant measures that can also benefit the overall performance of the engineering department; and
- create a built-in search functionality that can let the user search for a corresponding information which can save time.

C. Significance of the Study

The study is significant to the following entities:

Department Employees. this can let the employees such as the faculty body of the engineering department be aware that there is a more effective and technological way of managing their laboratory.

Laboratory Personnel (Technicians). this study aims to let the assigned laboratory personnel be more efficient with their work when it comes to handling the different information from the engineering laboratories.

Students. This study is also beneficial to the students since it lets them know that there is a more effective way of securing and processing their information when it comes to making transactions with their department's laboratory.

Researchers. This proposed system will benefit the researchers since it can let them expand their knowledge in the future whenever they are about to conduct another system.

Future Researchers. This research paper will be another benefit to any future researchers since they can use this as a guide and reference for their upcoming research study.

D. Scope and Limitations of the System

The researchers will develop a modernized Laboratory Management System for the College of Engineering at Benedicto's College which has:

Module Registration. This module is in charge of user registration such as the administrator.

Module Security. This module is in charge if the user is an authorized user or not.

Module Student Records. This module is for storing the student's information who will borrow the laboratory apparatus.

Apparatus Management. This module is intended for the inventory aspect of the laboratory apparatuses where they can be managed such as adding and updating their information.

Module Borrow Apparatus Records. This module is in charge of storing the borrowing records of the laboratory apparatuses.

Module Return Apparatus Records. This module is in charge of keeping the return records of the laboratory apparatuses after they are borrowed.

Module Penalty Records. This module is for monitoring and keeping the penalty records whenever a violation/penalty has been done.

However, the system is not capable of:

1. Providing live chat support.
2. Direct contact with the property custodian.
3. Reservation records.

CHAPTER II

Theoretical Background

A. Review of Related Systems

A Review of Related Systems is a systematic examination and comparison of existing software solutions that are relevant to the research topic. This involves identifying similar or related systems, analyzing their features and capabilities, and evaluating their strengths and weaknesses. The goal is to establish the current state of the field, identify potential gaps or limitations, and justify the need for the proposed system.

1. eLabNext

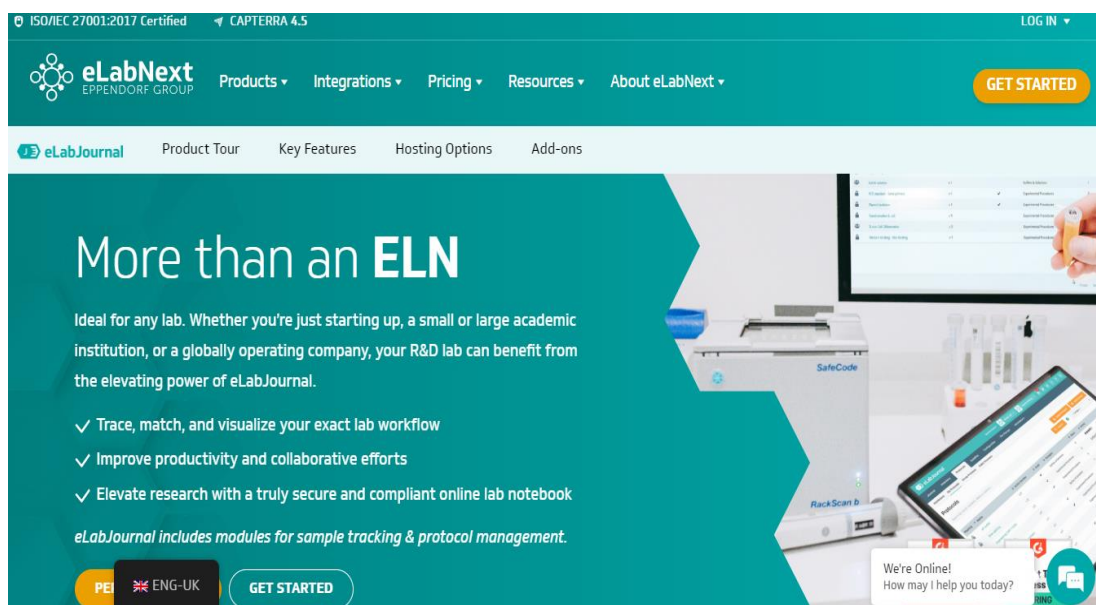


Figure 1: eLabNext Home Page

The eLab Journal website is an innovative online platform designed to simplify laboratory management and research documentation. Unlike traditional physical lab notebooks, eLab Journal

offers a comprehensive suite of features to enhance the efficiency and reliability of recording experimental data, managing project workflows, and ensuring compliance with regulatory standards. While it does not involve borrowing or returning like the proposed system, it encompasses features such as inventory management for tracking lab materials and equipment, albeit without traditional penalties associated with late returns. Its strengths lie in its ability to centralize and secure data, facilitate collaboration among researchers, and improve traceability of experiments. However, its weaknesses may include a learning curve for new users, potential dependency on internet connectivity, and subscription costs [1].

The proposed system will integrate more intuitive user interfaces, offline functionality, enhanced data analysis tools, and scalable pricing models to accommodate a wider range of research environments [1].

2. UBC Sauder: CLC Book and Equipment Borrowing

The screenshot displays the UBC Sauder myBCom website. At the top, the UBC Sauder School of Business logo is on the left, and navigation links for COOL, Canvas, David Lam Library, and SSC are on the right, along with a green search button. Below this is a dark blue header with the 'myBCom' logo and a horizontal menu with links: About UBC Sauder, Academic Success, Health & Wellness, Global Reach, Career & Experience, Courses, Money & Enrolment, and Support. A breadcrumb trail reads: Home > About UBC Sauder > Learning Commons and Library > Book and equipment borrowing. The main content area features a large green banner with the title 'CLC Book and Equipment Borrowing'. Below the banner, there are two main sections: 'Borrowing books' and 'Borrowing equipment'. The 'Borrowing books' section states that the Canaccord Learning Commons and David Lam Library have a variety of books, ebooks, and course reserves available to all students, and that all course reserves must be requested at the Service Desk. The 'Borrowing equipment' section states that UBC Sauder students are permitted to borrow items (for academic use only) from an equipment lending pool, located at the Service Desk, and that all equipment is lent on a first-come, first-served basis and a student card is required to sign items out. To the right of these sections, there is a 'Your Library card' section with a card icon, stating that the UBC card is the library card and is used for printing and copying at any UBC Library branch. Below this is a link to 'USING YOUR LIBRARY CARD'. At the bottom right, there is a 'Library Catalogue' section with a book icon and a link to 'Did you know you can borrow this...'. A footer line at the bottom states: 'Any questions about access, fees, or reservations can be directed to the Service Desk.'

Figure 2: UBC Sauder. CLC Book and Equipment Borrowing Homepage

The CLC Book and Equipment Borrowing website serves as a digital platform for users, primarily students and faculty, to access and borrow and return a variety of resources including textbooks, literature, and academic equipment. A significant strength of the website lies in its capacity to centralize access to educational materials, making it easier for users to find and borrow the items they need for their academic pursuits. It also potentially reduces the physical strain of managing and distributing these materials manually. However, it's not without its weaknesses; the user interface might be non-intuitive or outdated, making navigation difficult for some users. There might also be issues with inventory management, where the system fails to accurately reflect the availability of items, leading to frustrations or inconveniences [2].

The proposed system will address these weaknesses by incorporating a more user-centric and modern interface, ensuring that users of all tech-savviness levels can navigate the system with ease. It could also integrate a firmer system, real-time inventory management system, minimizing discrepancies between the online platform and actual availability. The proposed system will enhance security measures and privacy protocols is also being a focus, ensuring user data is protected. The proposed system would not only rectify existing shortcomings but also elevate the overall user experience, making the process of borrowing and returning equipment more efficient. This related system can greatly support the idea of the proposed system since it will act as an inspiration for the development of the proposed system [2].

3. Quartzzy

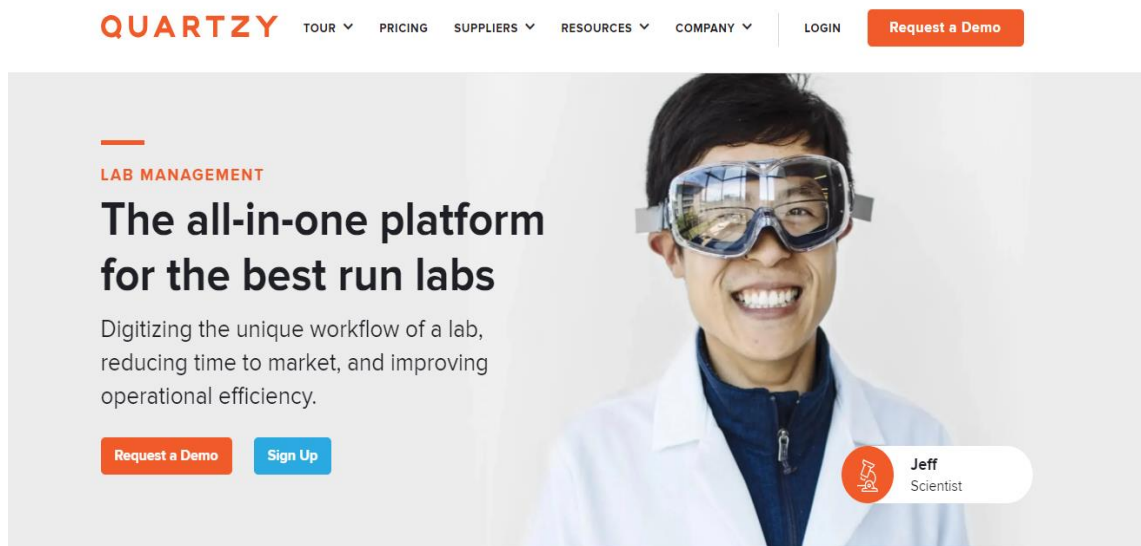


Figure 3: Quartzzy Homepage

Quartzzy is an online platform designed to streamline laboratory operations, offering a comprehensive suite of tools for inventory management, order tracking, and supply requests within the scientific community. While Quartzzy does not include features for borrowing or returning items like to the proposed system, nor does it implement penalties, it excels in facilitating efficient inventory oversight and reducing administrative burdens. Its strengths lie in its ability to centralize lab management tasks, enhance transparency in lab operations, and improve and communication among researchers [3].

However, its weaknesses may include a potential learning curve for new users and the absence of features catering to the unique needs of different lab environments. The proposed system aims to incorporate more intuitive user interfaces, customizable features for diverse scientific disciplines, and integrated borrowing and returning functionalities with automated penalty enforcement for overdue items. This approach could offer a more tailored and flexible solution,

accommodating a wider range of laboratory management needs and enhancing overall user satisfaction [3].

4. LabVantage

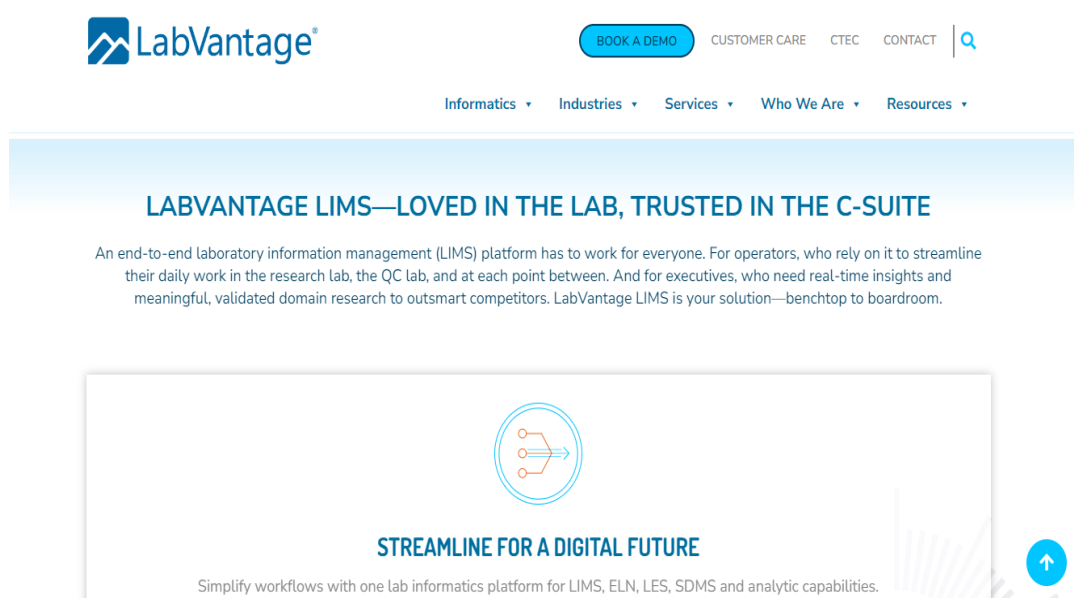


Figure 4: LabVantage Homepage

LabVantage Solutions, Inc. runs an extensive website highlighting its Laboratory Information Management System (LIMS) aimed at improving laboratory operations' efficiency and quality in various industries. Unlike the proposed system, it doesn't involve traditional borrowing, returning, or penalty features. Instead, it focuses on managing laboratory inventory, tracking samples, reagents, and other assets. Its strength lies in its flexibility and integration capabilities, fitting into diverse laboratory workflows and IT systems to enhance data integrity and operational efficiency [4].

However, some may see its complexity and need for significant configuration as drawbacks, especially for smaller labs with limited IT resources. The proposed system offers a more user-friendly interface with automated workflows, advanced analytics for predictive inventory management, and cost-effective solutions suitable for labs of all sizes [4].

5. LabArchives

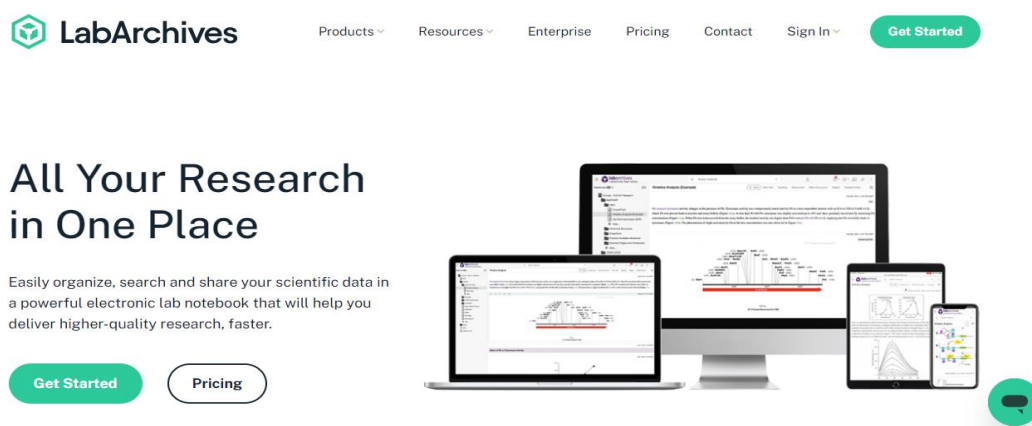


Figure 5: LabArchives Homepage

LabArchives is a web-based application designed for scientists to store, organize, share, and manage their laboratory data. It caters primarily to the needs of researchers, educators, and students, providing a secure platform for recording experiments, tracking results, and collaborating with team members. LabArchives has its ability to streamline the research process and ensure data integrity, making it a valuable tool in the scientific community [5].

Unlike traditional library systems, LabArchives does not operate on a borrowing or returning basis, and there are no penalties for its use. Instead, access is subscription-based, with various plans tailored to individual researchers, academic institutions, and corporate

entities. This model ensures users have constant access to their data and the suite of tools provided by LabArchives without worrying about overdue returns or penalties [5].

The strengths of LabArchives include its firm data management capabilities, collaborative features, compliance with research standards and regulations, and versatility across different scientific disciplines. However, it is not without its weaknesses. The interface is less intuitive than desired, and there can be a learning curve for those not familiar with electronic lab notebooks (ELNs). Additionally, the cost might be a barrier for individual researchers or smaller institutions [5].

The proposed system aims to enhance more user-friendly interface, incorporate seamless integration with a broader range of research tools and platforms could further enhance its attractiveness. Ultimately, the success of such a system would depend on its ability to meet the evolving needs of the scientific community while providing a seamless, efficient, and cost-effective research data management solution [5].

B. Matrix of Comparison

A matrix of comparison, often referred to as a comparison matrix, is a tool used to evaluate and contrast multiple options or variables against a set of criteria. It organizes information in a grid format, where rows typically represent the options or items being compared, and columns signify the criteria or factors considered important for the evaluation.

Systems/ Functions	eLabNext	UBC Sauder	Quartzy	LabVantage	LabArchives	Benedicto College of Engineering Laboratory Management
Registration	✓	✓	✓	✓	✓	✓
Responsiveness	✓		✓	✓	✓	✓
User-Friendly Interface	✓		✓	✓	✓	✓
Penalty Records		✓				✓
Borrowing Records		✓				✓
Returning Records		✓				✓
Apparatus Inventory List	✓		✓	✓		✓
Apparatus Referential Issuing						✓
Student Records						✓

Table 1: Matrix of Comparison of the Review of Related Software.

C. Definition of Terms

The Definition of terms is vital in academic and professional writing to clarify key concepts, minimize ambiguity, and enhance reader comprehension. Definitions can be sourced from dictionaries, context, or author interpretation to convey precise meanings, improving communication effectiveness.

External Factor. An influence or element that originates outside of the particular system that can affect performance and operations.

Apparatus. A collection of different equipment, instruments, or tools that are designed to do a specific task or to fulfill a specific purpose, they are commonly used to achieve objectives efficiently.

Penalty Records. These are records that are being kept regarding the different penalties that have been done by any individuals or entities for various reasons.

Laboratory Personnel. These are the assigned working individuals for the Laboratories, they conduct different tasks such as helping in managing the laboratory as well as monitoring the well-being of the laboratory apparatus.

Property Custodian. They are in charge of safeguarding the status of the assets or equipment.

Foreign Damage. This is the kind of damage or harm that is caused by other issues that can greatly affect the quality, usability, and integrity of the products.

CHAPTER III

Methodology

A. Research Design

a. Research Environment

This research takes place at Benedicto College, Mandaue Campus, Cebu City specifically within the College of Engineering Department. The Department is currently being handled by their college dean, Dr. Nimfa O. Rodriguez. The department currently has an approximate of 12 faculty members alongside 2 laboratory personnel with 3 more that will be starting in the next semester. The College Dean and the laboratory personnel are the ones in charge of managing the laboratories since they keep track of the apparatuses and requirements, handling the records monitoring with the borrowing and returning as well as the penalty records.



Figure 6: Benedicto College, Mandaue Campus



Figure 7: Engineering Department

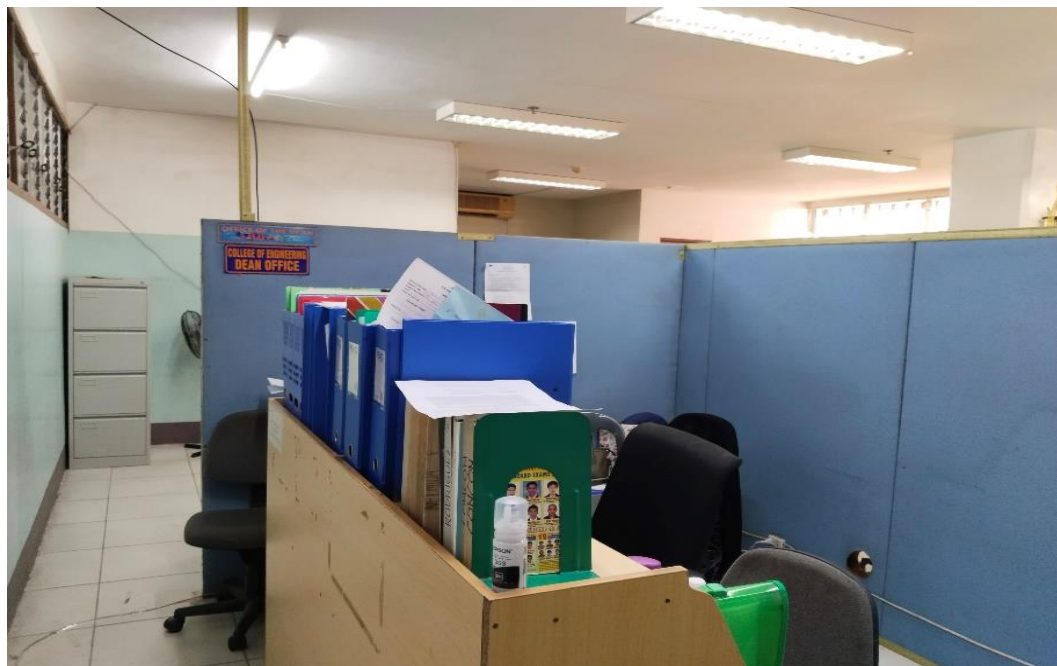


Figure 8: Engineering Department



Figure 9: Laboratory



Figure 10: Laboratory Apparatuses

b. Research Map

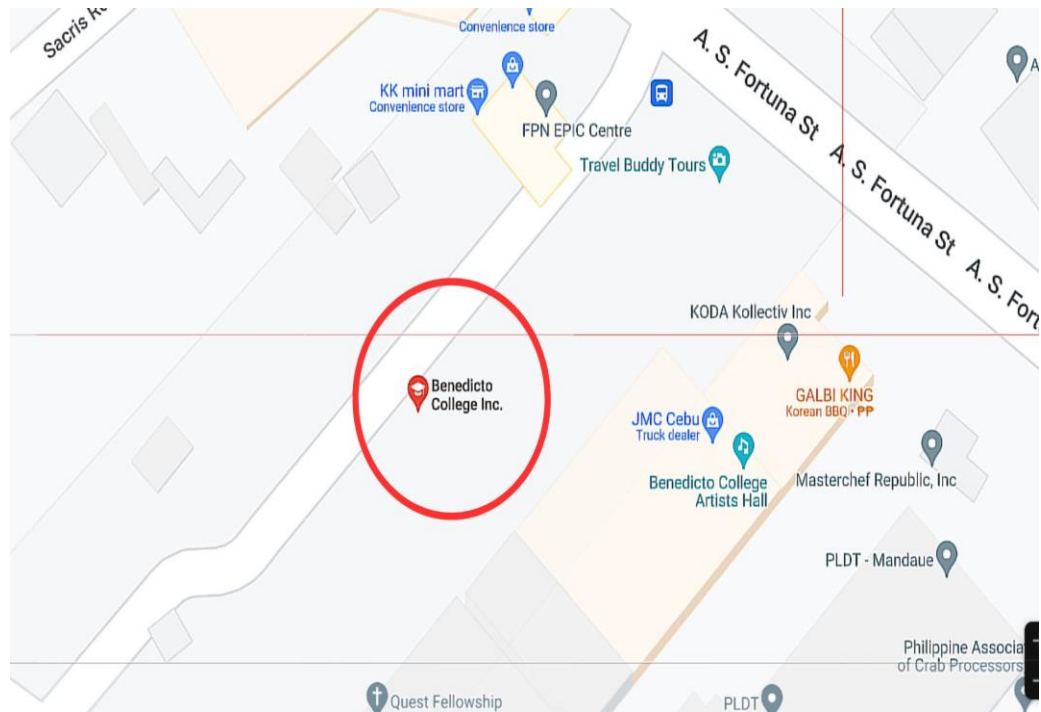


Figure 11: Google Maps Overview of Benedicto College, Mandaue Campus

c. Methods and Techniques

The researchers will use the applied research approach in order to address and identify the different problems and pain points in the research environment. This is done by collecting and analyzing the data in order to systematically create its solutions. The following are the methods and techniques that will be used throughout this research for the proposed system.

Rapid Application Development (RAD).

System Prototyping. This is to create an initial simplified version of the system in order to demonstrate its functionalities and to gather client or user feedback. This method is a good

and flexible way of developing the system because it helps in refining and validating the system design and requirements so that it can meet the user's satisfaction.

Interview Technique. This elicitation technique will be used by the researchers in order to gather in-depth information and data for the proposed system.

d. Models

The researchers will use different conceptual model diagrams to represent and apply the flow of the proposed system:

Use Case Model. The researchers will use the use case model to describe how the users will interact with the system and can also identify their different roles.

Class Model. The researchers will also apply the usage of the class model diagram to show the different data models of the proposed system.

Entity Relationship Model. More importantly, the researchers will also use the Entity-relationship model diagram to create a visual representation of various "entities" such as people, objects, or concepts and to relate to each other within the system.

e. Tools

The following are the tools that will be used by the researchers in developing the proposed system:

Visual Studio Community. The main Integrated development environment (IDE) that will be used for the development of the proposed system.

Microsoft SQL Server Management Studio (SSMS). This is used to manage and create the database infrastructure.

MS Word. The researchers will use this application to create the documentation of the proposed system.

Figma. This will be used to create and design the initial prototype of the proposed system.

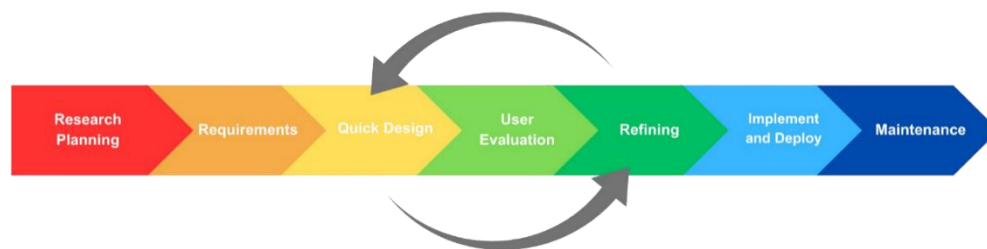
GitHub. This will be used in storing the important extension files of the system during the development/implementation phase, this is used for version tracking whenever the system needs emergency updates or backup.

Microsoft Edge. The researchers will use this for the testing of the system as it is easily compatible with the debugger of Visual Studio Community.

Opera GX. The researchers will use this browser to browse additional information that can be used for the development of the system.

f. Development Process

The table diagram shown below is the development process with the appliance of the Rapid Application Development methodology. The researchers started their Research Planning during the month of February and have further conducted their requirements gathering from the 4th week till March. During the third week of April until July, the researchers will start to apply the System Prototyping method alongside increasing the functional system which will consume more time until it can be fully functional by the month of November.



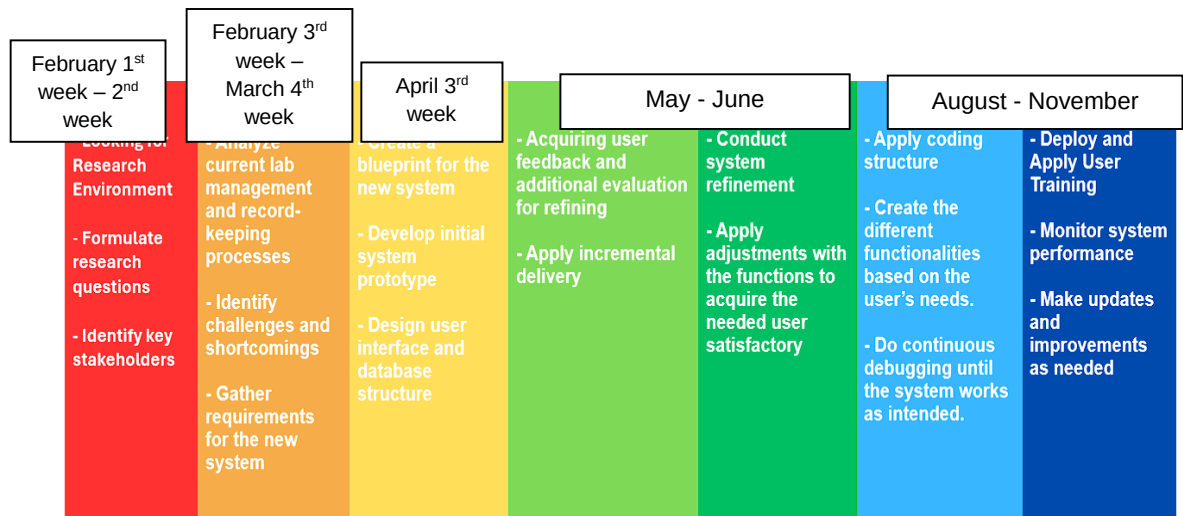


Figure 12: Development Process of Laboratory Management System

B. Requirements Analysis

a. Functional Requirements

The functional requirements represent what the system can do with the following data.

It tackles the data that needs to be inputted for it to be processed to get a certain output:

Input

The following are the inputs:

- Administrator registration and login details
- Student data
- Apparatus inventory data
- Input penalty data
- Input Borrower and Return data
- Input Search data

Process

The following are the processes:

- Process user registration
- Process user authentication
- Process adding, updating, or removing student information
- Process adding, or updating information from the apparatus inventory list
- Process adding, or updating penalty record data
- Process adding, or updating the apparatus borrow-return records data
- Process search query that will try and match the search data input

Output

The following are the outputs:

- Granting access to the main page when the authentication process is completed
- Updated student information layout
- Updated apparatus inventory list
- Updated penalty records information
- Updated Borrow and Return records information
- Search results based on the specific input data

b. Non-Functional Requirements

The non-functional requirements will refer to how the system will respond or performs attributes. This is how the system will behave when it is being used by the user.

- The system will store the registered user information in a secured database system.
- The system will authenticate the user's access by making sure it matches the corresponding username and password correctly as well as making sure that the registered credentials are inputted properly.

- The system will apply encryption when storing the user's password in order for it to be secured.
- The system will try and match the search data input from the user to display the relevant information on what is being searched.
- The system will update the corresponding information from the students, inventory, borrowing-returning records, and penalty records as soon as the administrator makes any updates.
- The system will provide a user-friendly interface for the administrator to have easy navigation and accessibility.
- The system will display an error indication if the user searches for a specific information that does not match any records from the system.
- The system will perform proper issuing techniques in order to make an accurate and secure borrowing transaction for the laboratory apparatus.

c. Usability Requirements

A usability requirement describes how the system should be operated by the user according to their characteristics.

- The user should possess a basic understanding of computer operations and be familiar with common computer functionalities such as using a keyboard and mouse, navigating through menus, and managing records.
- The user should possess a clear understanding of the operation and functionality of the proposed system.

d. Use Case Model

This use case model demonstrates how the College of Engineering Laboratory Management System allows users to interact with the system. The diagram includes two key actors, the Admin and Staff (Laboratory Personnel), and it outlines the functionalities available to them.



Figure 13: Business Use Case of College of Engineering Laboratory Management System.

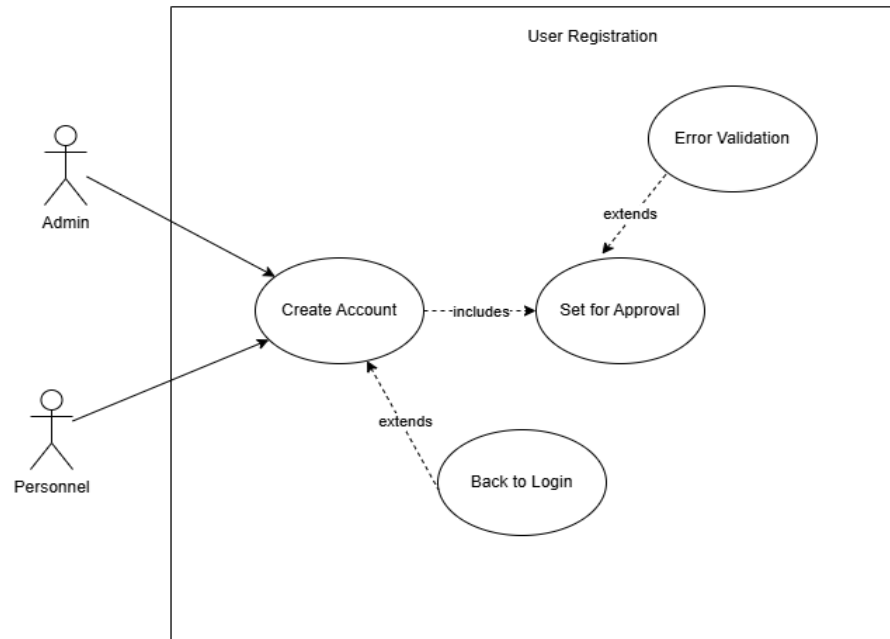


Figure 14: System Use Case of User Registration.

Use Case Name: Create Account

Purpose: To enter the user registry data and login credentials.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to create an account in order to use the system.

Post-Condition: The user has entered the needed information with their login credentials.

Steps:

USER	SYSTEM
1. User clicks on Registration 3. User will now fill-up the needed input registration.	2. System loads the Register

Use Case Name: Error Validation

Purpose: This is to check any exceptions encountered during the registration.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user enters some data without double checking its format or values.

Post-Condition: The system will display an error message indicating the error exception.

Steps:

USER	SYSTEM
1. The user enters their registration data.	2. If the system checks that the email has not been verified 2.1 Display Email Authentication error message. 2.2 else if there are blank input fields. 2.3 Display Error Message. 2.4. else if none, Continue.

Use Case Name: Back to Login

Purpose: This is when the user wants to go back to the login page.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to return to the previous login page.

Post-Condition: The user will be redirected back to the login instance.

Steps.

USER	SYSTEM
1. The user clicks on the "Back to Login button".	2. The register instance will close and will redirect the user back to the Login form.

Use Case Name: Set for Approval

Purpose: This is where the registered user is set for their approval of registry.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user has already entered their credentials and their email has been verified.

Post-Condition: An assigned admin will enter the following credentials in order to approve or disapprove their registry.

Steps:

USER	SYSTEM
1. The user presses on the Registry button. 3. An authorized admin will enter the needed verification inputs on the spot.	2. The system will display the Admin Approval form. 4. If there is an authorized input 4.1 Registration approved and success 4.2 else the Admin may reject the registration.

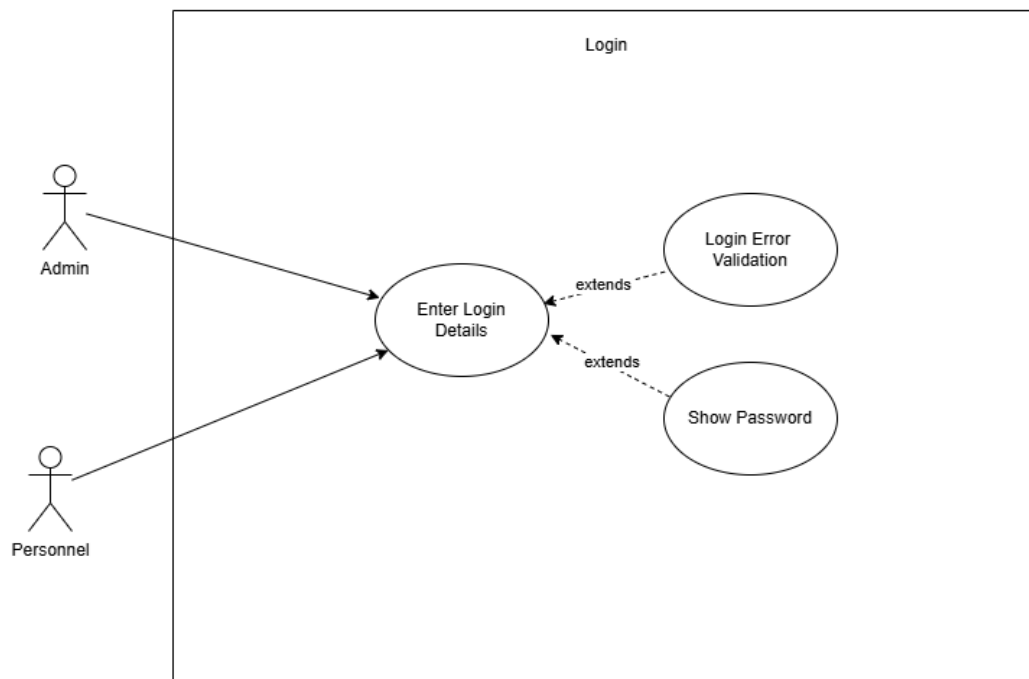


Figure 15: System Use Case of Login.

Use Case Name: Enter Login Details

Purpose: This is where the user will enter their username and password.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to login to the system.

Post-Condition: The user has entered their login credentials.

Steps:

USER	SYSTEM
1. The user will load the system.	2. The system will display the Login form.
3. The user will enter their username and password.	

Use Case Name: Login Error Validation

Purpose: This is to validate any incorrect user login credentials.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user enters an incorrect username or password.

Post-Condition: A message box will appear indicating the login error.

Steps:

USER	SYSTEM
1. The user will enter their login credentials and clicks on Login.	2. If the username or password does not match 2.1 Display Error Message. 2.2 else if there are blank fields 2.3 Display an error message indicating blank field.

Use Case Name: Show Password

Purpose: This is the option where the user wants to see and check their password during login.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to see or check their password.

Post-Condition: The user can see their password.

Steps:

USER	SYSTEM
1. User clicks on the "Show Password" checkbox.	2. System will disable the PasswordChar and displays the complete string.

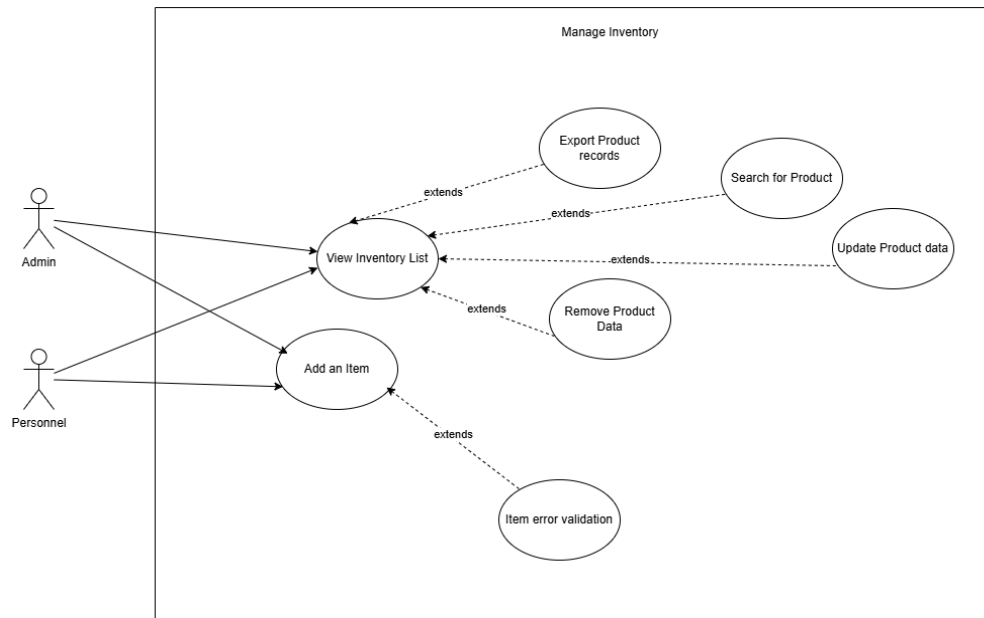


Figure 16: System Use Case of Manage Inventory.

Use Case Name: View Inventory List

Purpose: The user will view the Inventory List.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition. The user wants to access and view the inventory list.

Post-Condition: The system will display the inventory list.

Steps:

USER	SYSTEM
1. The user selects “View Apparatus List”.	2.The system will check if they match.

Use Case Name: Export Product records

Purpose: To export the system's data into Excel.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to export all the records in a worksheet file.

Post-Condition: The system will export its following records into Excel.

Steps:

USER	SYSTEM
1. The user clicks on the "Export" button.	2. The system will export all the records from the system into an Excel file.

Use Case Name: Search for Product

Purpose: To search a product based on its details.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to search for a specific product.

Post-Condition: The system will display the product based on the input data on the search textbox.

Steps:

USER	SYSTEM
1. The user will type in the Search bar or do a filter search.	2.if the data or record exists 2.1 The system will display the searched results 2.2 else, display blank dataset or error message, 3. Proceed again to step 1 for optional search.

Use Case Name: Update Product Data

Purpose: To update the product details.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to update the product details.

Post-Condition: The system will display the updated product details.

Steps:

USER	SYSTEM
1. The user will click on a content row to check.	
	2.The system will bind the values into their following controls.
3. The user will edit or update some of its information.	
4. The user clicks on Update.	
	5. The system will update the records based on the changes.

Use Case Name: Remove Product Data

Purpose: To remove a product from the inventory.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user wants to remove the product.

Post-Condition: The system will remove the product from the inventory list.

Steps:

USER	SYSTEM
1. The user will click on a row of records.	
	2.The system will then bind them into their corresponding controls.
3. The user will press on the “Delete” button.	
	4. The system will delete the corresponding selected records.

Use Case Name: Add an Item

Purpose: The user will add an item to the inventory.

Triggering Actor: Admin/Personnel

Benefitting Actor:Admin/Personnel

Pre-Condition:The user wants to add an item in the inventory.

Post-Condition:The system will read and add the item details in its inventory records list.

Steps:

USER	SYSTEM
1. The user will click on the “Add Apparatus” Menu Item.	
	2.The system will display the “AddApparatus” instance.
3. The user will input the item details and clicks add.	
	4. The system will add the item to its records.

Use Case Name: Item Error Validation

Purpose: To validate the item's details.

Triggering Actor: Admin/Personnel

Benefitting Actor: Admin/Personnel

Pre-Condition: The user enters the item details.

Post-Condition: The system will generate an error message based on the item detail exception error.

Steps:

USER	SYSTEM
1. The user will enter the item details and clicks on "Add".	2.The system will check if all the entered values are valid. 3. if the data is invalid 3.1 display an error message and proceed to step 1 3.2 else continue

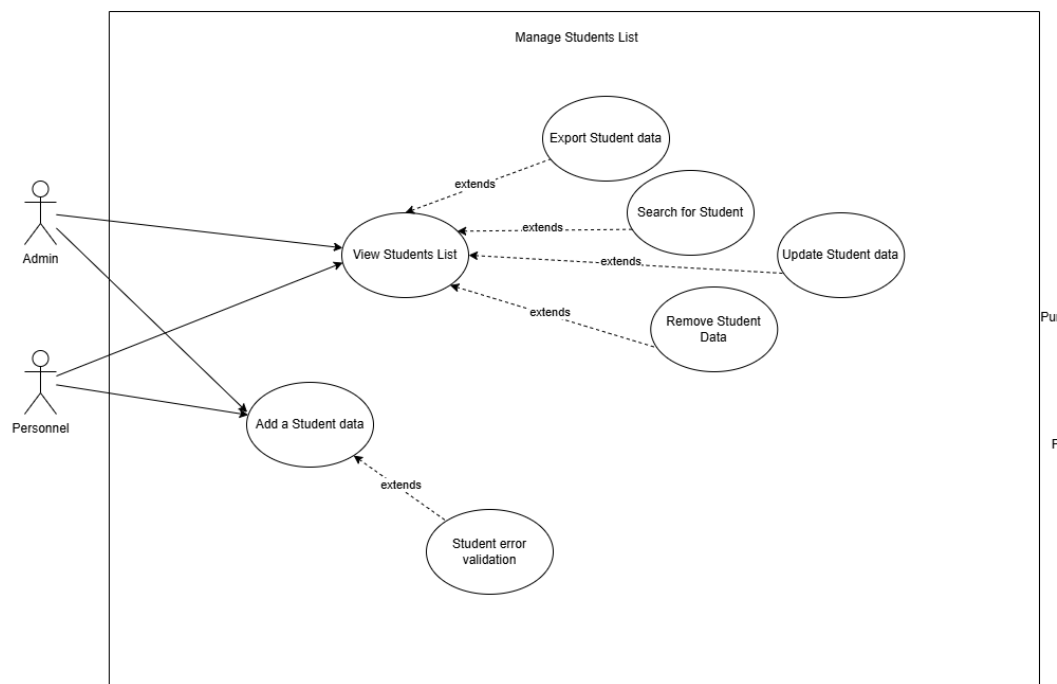


Figure 17: System Use Case of Manage Student List.

Use Case Name: View Students List

Purpose: To display a list of students to authorized users.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to view the student lists.

Post-Condition: Student data is displayed.

Steps:

USER	SYSTEM
1. The user will click on the "View Students List" Menu selection.	2. The system will display the Student List form instance.

Use Case Name: Export Student Data

Purpose: To allow authorized users to export student data in a worksheet.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to export the records.

Post-Condition: Student data is exported in the requested format.

Steps:

USER	SYSTEM
1. The user will click on the Export records button.	2.The system will export all the student records from the student list into an Excel or worksheet file.

Use Case Name: Search for Student

Purpose: To allow authorized users to search for specific student information.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to search for a student information.

Post-Condition: Search results are displayed.

Steps:

USER	SYSTEM
1. The user will type on the search bar or apply filter search.	2. if the data or record exists 2.1 System will update the display 2.2 else display null dataset or an error message 2.3 else proceed back to step 1.

Use Case Name: Update Student Data

Purpose: To allow authorized users to modify student information.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to update the student data.

Post-Condition: Student data is updated.

Steps:

USER	SYSTEM
1. The user will click on the student records.	
	2. The system will bind all data into their corresponding controls.
3. The user will update some of its student information.	
4. The user will click on Update.	
	5. The system will update the student's corresponding record.

Use Case Name: Remove Student Data

Purpose: To allow authorized users to delete student records.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to remove the student data.

Post-Condition: Student data is deleted.

Steps:

USER	SYSTEM
1. The user will click on student's records.	
	2. The system will then bind their records into their controls.
3. The user will click on the "Delete" button.	
	4. The system will delete the student's records and will update the list.

Use Case Name: Add a Student Data

Purpose: To allow authorized users to create new student records.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: The user wants to add a student data.

Post-Condition: New student data is added.

Steps:

USER	SYSTEM
1. The user will click on the “Add a Student” from the Menu.	
	2. The system will display the “Add Students” instance.
3. The user will input the student’s data and will click “Add”.	
	4. The system will add the student data into its records.

Use Case Name: Student Error Validation

Purpose: To validate the accuracy and completeness of student data entered into the system.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: Student data is being entered or updated.

Post-Condition: Errors in student data are identified and reported to the user.

Steps:

USER	SYSTEM
1. The user will input the student's details and will clicks on "Add".	2. if the inputs are not correct or blank 2.1 Display an error message 2.2 else proceed back to step 1.

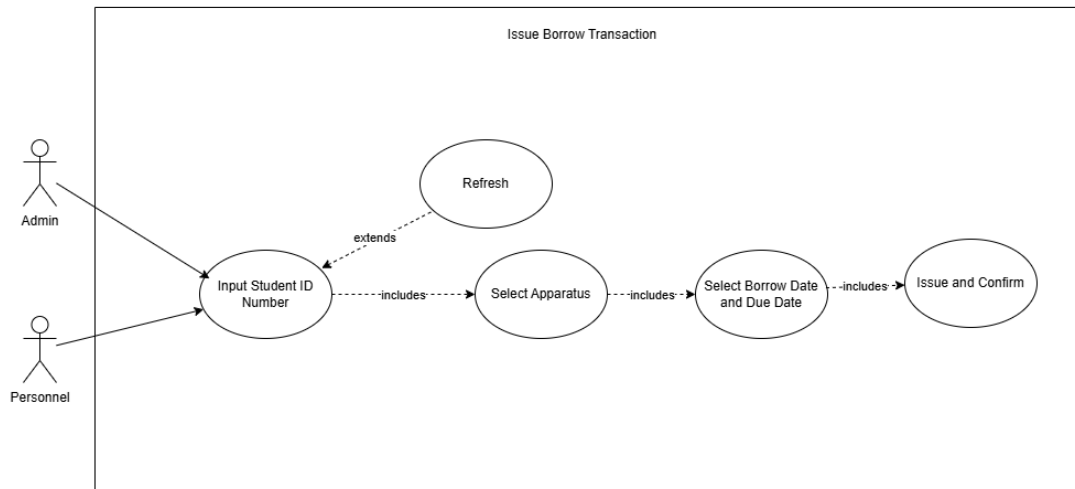


Figure 18: System Use Case of Issue Borrow Transaction.

Use Case Name: Input Student ID Number

Purpose: To search the student's ID number.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a borrow transaction.

Post-Condition: Student ID number is entered into the system.

Steps:

USER	SYSTEM
1. The user will click on the “Borrow Transaction” Menu Item. 3. The user will type the student ID Number.	2.The system will load the Borrow Transaction from instance. 4. The system will check if the student exists on the student list 5. if the ID- Number exists 5.1 Continue the transaction 5.2 else the user will proceed back to step 3.

Use Case Name: Refresh

Purpose: To update the displayed information or reset the input fields.

Triggering Actor: User

Benefitting Actor: User

Pre-Condition: User is interacting with the system.

Post-Condition: The displayed information or input fields are refreshed.

Steps:

USER	SYSTEM
1. The user will press on the Refresh button.	2.The system will refresh its instance back to its default display.

Use Case Name: Select Apparatus

Purpose: To choose the specific item to be borrowed.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a borrow transaction.

Post-Condition: The selected apparatus name is recorded.

Steps:

USER	SYSTEM
1. The user will select an apparatus for the borrow transaction. 3. The user will proceed to select the items and their quantities.	2.The system will select the items from the inventory with its available quantities.

Use Case Name: Select Borrow Date and Due Date

Purpose: To determine the borrowing period for the item.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a borrow transaction.

Post-Condition: Borrow and due dates are selected.

Steps:

USER	SYSTEM
1. The user will select the borrow date and due date.	2.The system will display the corresponding date-time picker with its format.

Use Case Name: Issue and Confirm

Purpose: To finalize the borrow transaction and issue the item to the student.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel, Student

Pre-Condition: All necessary information has been entered and validated.

Post-Condition: The borrow transaction is confirmed, and the item is issued to the student.

Steps:

USER	SYSTEM
1. The user will select the borrow date and due date. 2. The user clicks on the Confirm button.	3. The system will check for errors 4. if the dates or the input data are correct 4.1 The system will accept and issue the borrow transaction 4.2 else, display an error message and proceeds back to step 1.

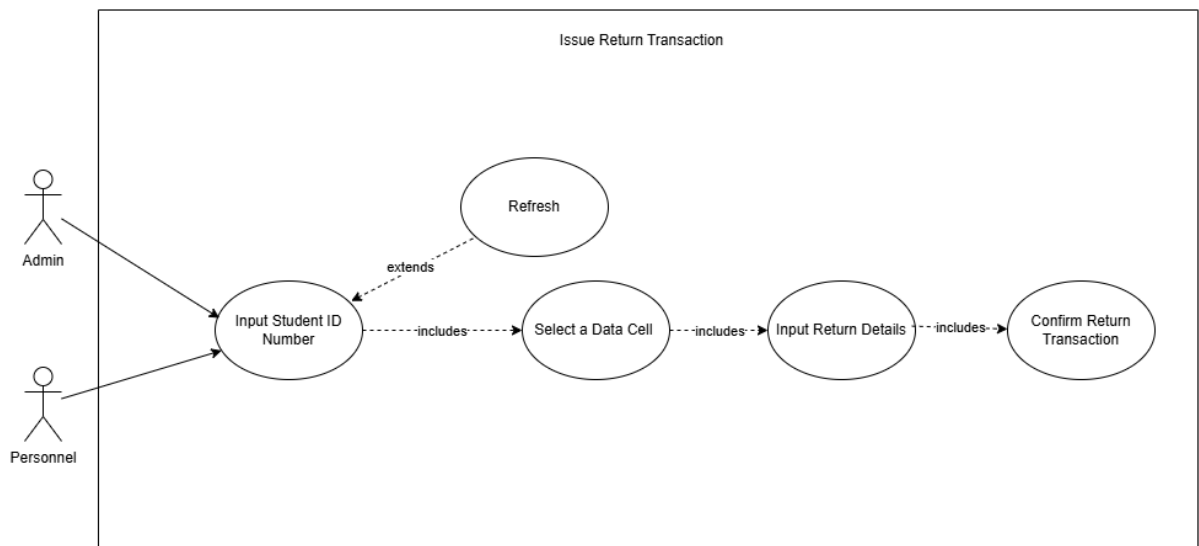


Figure 19: System Use Case of Issue Return Transaction.

Use Case Name: Input Student ID Number

Purpose: To capture the student's identification information.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a return transaction.

Post-Condition: Student ID number is entered into the system.

Steps:

USER	SYSTEM
1. The user clicks on the Return Transaction from the menu strip. 3. The user will then type a student ID number on the search bar.	2. The system will display the Return Transaction instance. 4. The system will search for the student with a borrow transaction. 5. if the transaction exists, the system will update its records display. 5.2 else, the system will display an empty records list or an error message. 5.3 else, proceed back to step 1.

Use Case Name: Refresh

Purpose: To update the displayed information or reset the input fields.

Triggering Actor: User

Benefitting Actor: User

Pre-Condition: User is interacting with the system.

Post-Condition: The displayed information or input fields are refreshed.

Steps:

USER	SYSTEM
1. The user will press on the Refresh button.	2. The system will refresh the Return Transaction form and will return to its default display.

Use Case Name: Select a Data Cell

Purpose: To select the specific borrowed item to be returned.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a return transaction.

Post-Condition: The selected data cell will bind the Borrow Transaction record

Steps:

USER	SYSTEM
1. The user will click on a data cell.	2. if the data cell is not empty 2.1 Bind the corresponding records to its controls. 2.2 else, the system will display an error message and proceeds back to step 1.

Use Case Name: Input Return Details

Purpose: To record the return date and any relevant comments or observations.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a return transaction.

Post-Condition: The Return Details are entered.

Steps:

USER	SYSTEM
1. The user will input the return date, remarks, and quantity.	2. The system will read the inputted date and numeric value and also the remark value.

Use Case Name: Confirm Return Transaction

Purpose: To finalize the return process and update the system records.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel, Student

Pre-Condition: All necessary information has been entered and validated.

Post-Condition: The return transaction is confirmed, and the item's status is updated to
"returned."

Steps:

USER	SYSTEM
1. The user will finalize the inputted data. 2. The user will click on Confirm.	3. The system will check if the return details are correct. 4 if details are correct 4.1 Proceed to issue return and completes the transaction. 4.2 else, display on error log message and the user will proceed back to step 1.

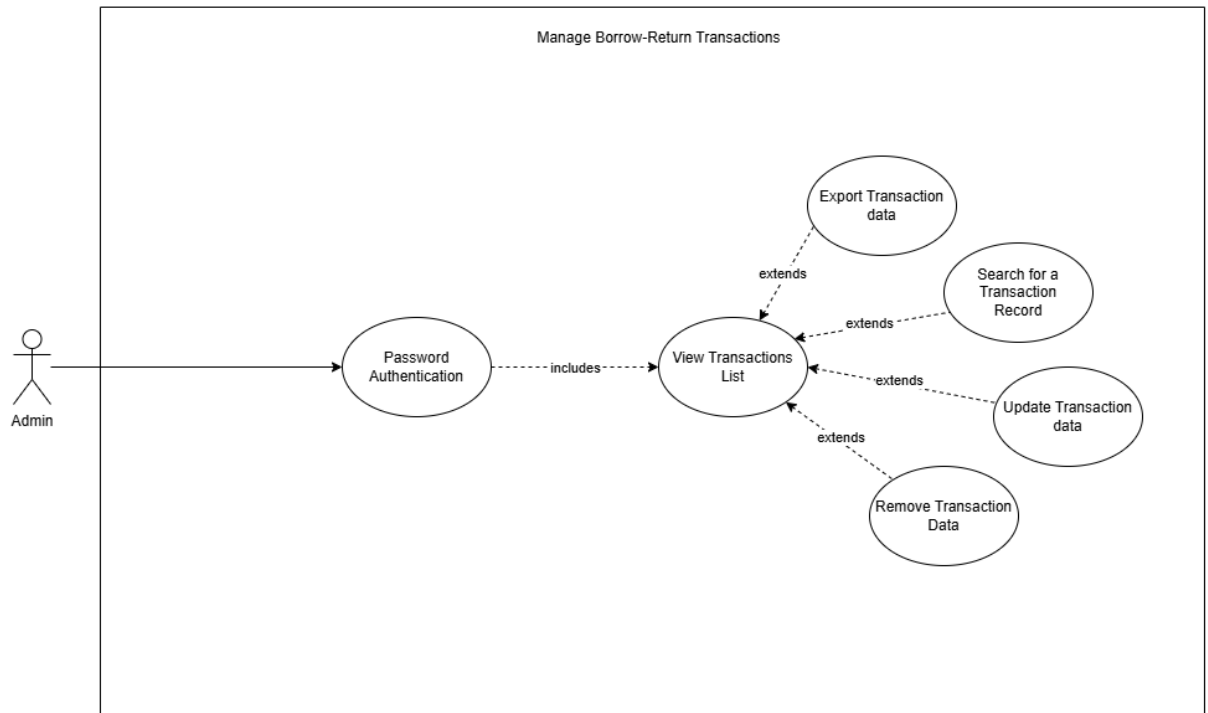


Figure 20: System Use Case of Manage Borrow-Return Transactions.

Use Case Name: Password Authentication

Purpose: To verify the user's identity through password authentication.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User is attempting to access the transaction records.

Post-Condition: User's identity is verified.

Steps:

USER	SYSTEM
1. The user will click on the Manage Transactions from the Menu strip. 3. The user will input their corresponding account password.	2. The system will open a security check instance. 4. The system will verify if the password matches 5. if the password match 5.1 proceeds to Transaction management 5.2 else, the system will reject the access and the user will proceed back to step 3.

Use Case Name: View Transactions List

Purpose: To display a list of all borrow-return transactions.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Transactions list is displayed.

Steps:

USER	SYSTEM
1. The user has now been verified. 2. The user proceeds.	3. The system will display the transaction records instance.

Use Case Name: Export Transaction Data

Purpose: To allow authorized users to export transaction data in a specific format.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Transaction data is exported in the requested format.

Steps:

USER	SYSTEM
1. The user will click on the Export to Excel Button.	2. The system will export all the records into a worksheet file.

Use Case Name: Search for a Transaction Record

Purpose: To allow authorized users to search for specific transaction information.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Search results are displayed.

Steps:

USER	SYSTEM
1. The user type on the search bar.	2. if the records or data exist 2.1 the system will update the dataset display. 2.2 else, the system will display a blank data set or an error message. 2.3 The user proceeds back to step 1.

Use Case Name: Update Transaction Data

Purpose: To allow authorized users to modify transaction information.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Transaction data is updated.

Steps:

USER	SYSTEM
1. The user type on the search bar. 3. The user will update the transactions details. 4. The user clicks on the update button.	2. The system will bind all its records on their corresponding controls. 5. if the inputted updates are correct 5.1 The system will update its transaction records. 5.2 else, an error message will be displayed and the user will be back to step 3.

Use Case Name: Remove Transaction Data

Purpose: To allow authorized users to delete transaction records.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Transaction data is deleted.

Steps:

USER	SYSTEM
1. The user will select a data cell.. 3. The user will click on the Delete button.	2. The records will be binded in its corresponding controls. 4. The system will delete the corresponding transaction records.

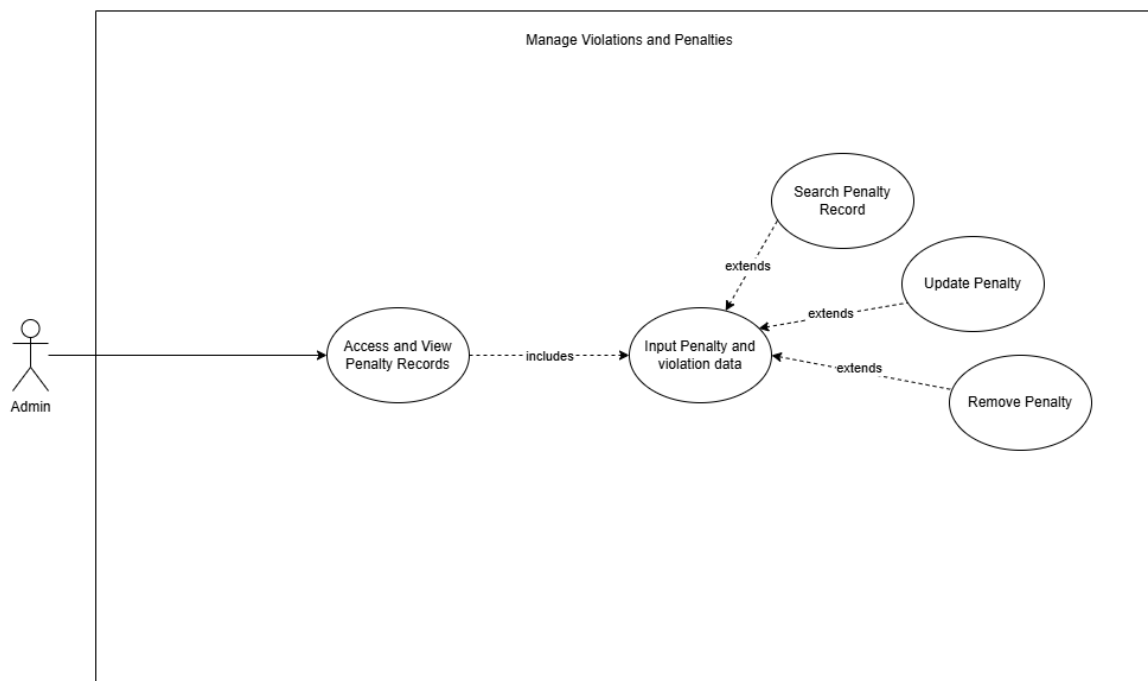


Figure 21: System Use Case of Manage Violations and Penalties.

Use Case Name: Access and View Penalty Records

Purpose: To allow authorized users to view and manage penalty records.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Penalty records are displayed.

Steps:

USER	SYSTEM
1. The user will click in the Penalties and Violations menu item.	2. The system will display the Penalty Records instance.

Use Case Name: Input Penalty and Violation Data

Purpose: To record new violation and penalty information.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Violation and penalty data is entered into the system.

Steps:

USER	SYSTEM
1. The user will input the transaction ID reference. 3. The user will then input the penalty data and the violation data. 4. The user will finalize the input data. 5. The user will click on the "ADD" button.	2. The system will automate the existing details from the transactions. 6. The system will check for any error handling. 7. if the input data are correct 7.1 proceed to add the penalty record 7.2 else, display and error message and the user will be back to step 4.

Use Case Name: Search Penalty Record

Purpose: To allow authorized users to search for specific penalty records.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: The search result is displayed.

Steps:

USER	SYSTEM
1. The user will type on the Penalty Search Bar.	2. If the record exists 2.1 Displays the search result 2.2 else, display a blank dataset and the user will proceed back to step 1.

Use Case Name: Update Penalty

Purpose: To allow authorized users to modify penalty information.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Penalty data is updated.

Steps:

USER	SYSTEM
1. The user will click on a Penalty data cell.	2. The records from the cells will be binded on the controls.
3. The user inputs some updates at the penalty records.	
4. The user will finalize the update inputs.	
5. The user will click on the update button.	6. If the update inputs are valid
	6.1 The system will update the records.
	6.2 else, displays an exception error log and the user proceeds back to step 1.

Use Case Name: Remove Penalty

Purpose: To allow authorized users to delete penalty records.

Triggering Actor: Admin

Benefitting Actor: Admin

Pre-Condition: User has appropriate permissions.

Post-Condition: Penalty data is deleted.

Steps:

USER	SYSTEM
1. The user will click on the Penalty Record.	2. The system will bind the records to its control.
3. The user will click the “Delete” button.	4. The system will delete the corresponding records that was selected.

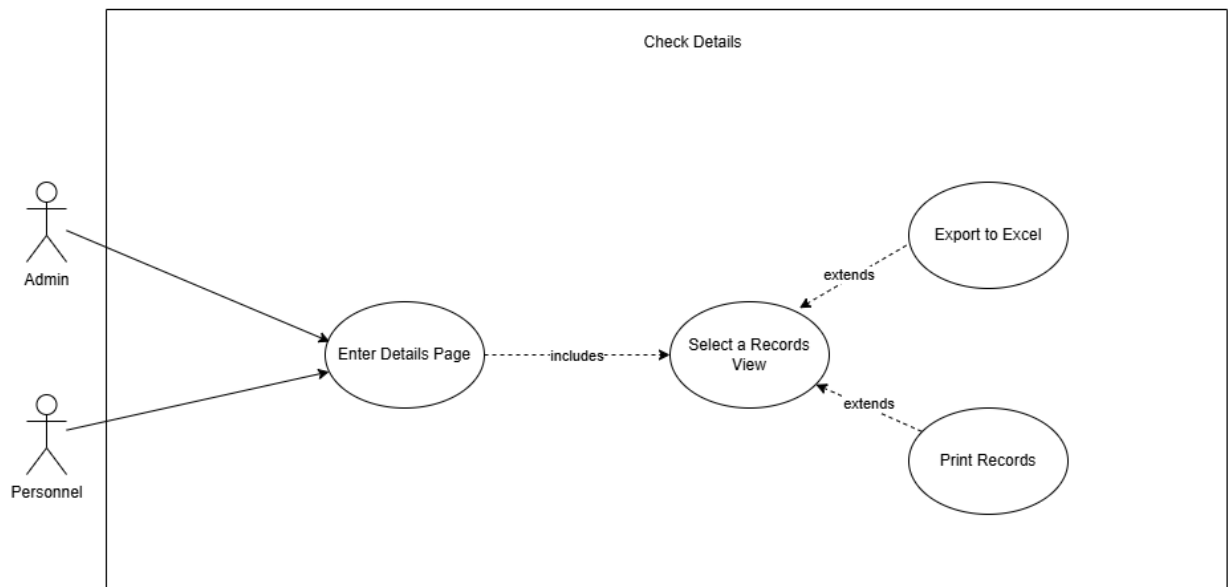


Figure 22: System Use Case Check Details.

Use Case Name: Enter Details Page

Purpose: To allow users to input specific criteria or parameters for the report.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is initiating a report generation process.

Post-Condition: Report details are entered.

Steps:

USER	SYSTEM
1. The user will click on the “View Details” menu item.	2. The system will display the details instance form.

Use Case Name: Select a Records View

Purpose: To choose the specific data set to be included in the report.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: User is generating a report.

Post-Condition: The selected records view is determined.

Steps:

USER	SYSTEM
1. The user will select on the combo box which records to view.	2. If the user selects "Inventory List" 2.1 The system displays the inventory records 2.2 else if the user selects the "student list" 2.3 The system will display the student list records 2.4 else if the user selects "Borrowed Apparatus" 2.5 The system will display the "Borrow records" 2.6 else if the user selects the "Return Apparatus" 2.7 The system will display the return records. 2.8 else the instance remains by default.

Use Case Name: Export to Excel

Purpose: To export the generated report data to an Excel spreadsheet.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: Report is generated.

Post-Condition: Report data is exported to Excel.

Steps:

USER	SYSTEM
1. The user will click on the Export button.	2. The system will export the corresponding records into a worksheet file.

Use Case Name: Print Records

Purpose: To apply a predefined report template and print the formatted report.

Triggering Actor: Admin, Personnel

Benefitting Actor: Admin, Personnel

Pre-Condition: Report is generated and a template is selected.

Post-Condition: Report is printed using the selected template.

Steps:

USER	SYSTEM
1. The user will click or press the Print Button.	2. The system will place the records in a scaffold template. 3. The system will then allow the printer to print the records.

e. Class Model

The Class Model represent the graphical structure of our system and it showsthe relation between the objects and the classes of our system.

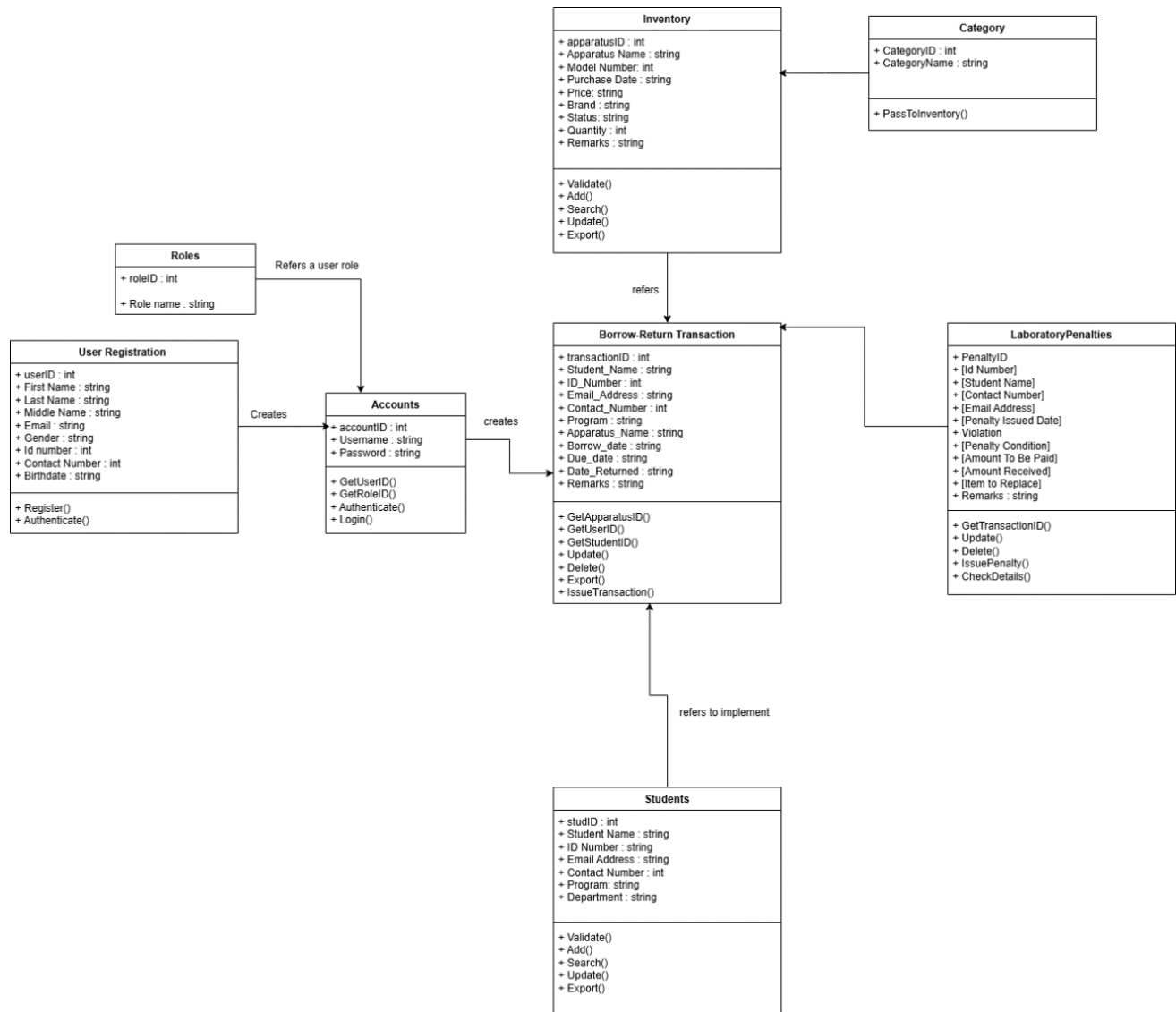


Figure 23:Class Model

f. Entity Relationship Diagram

The following figure shows the Entity-Relationship Diagram (ERD), which depicts the relationship between database entity sets and their attributes stored in a database.

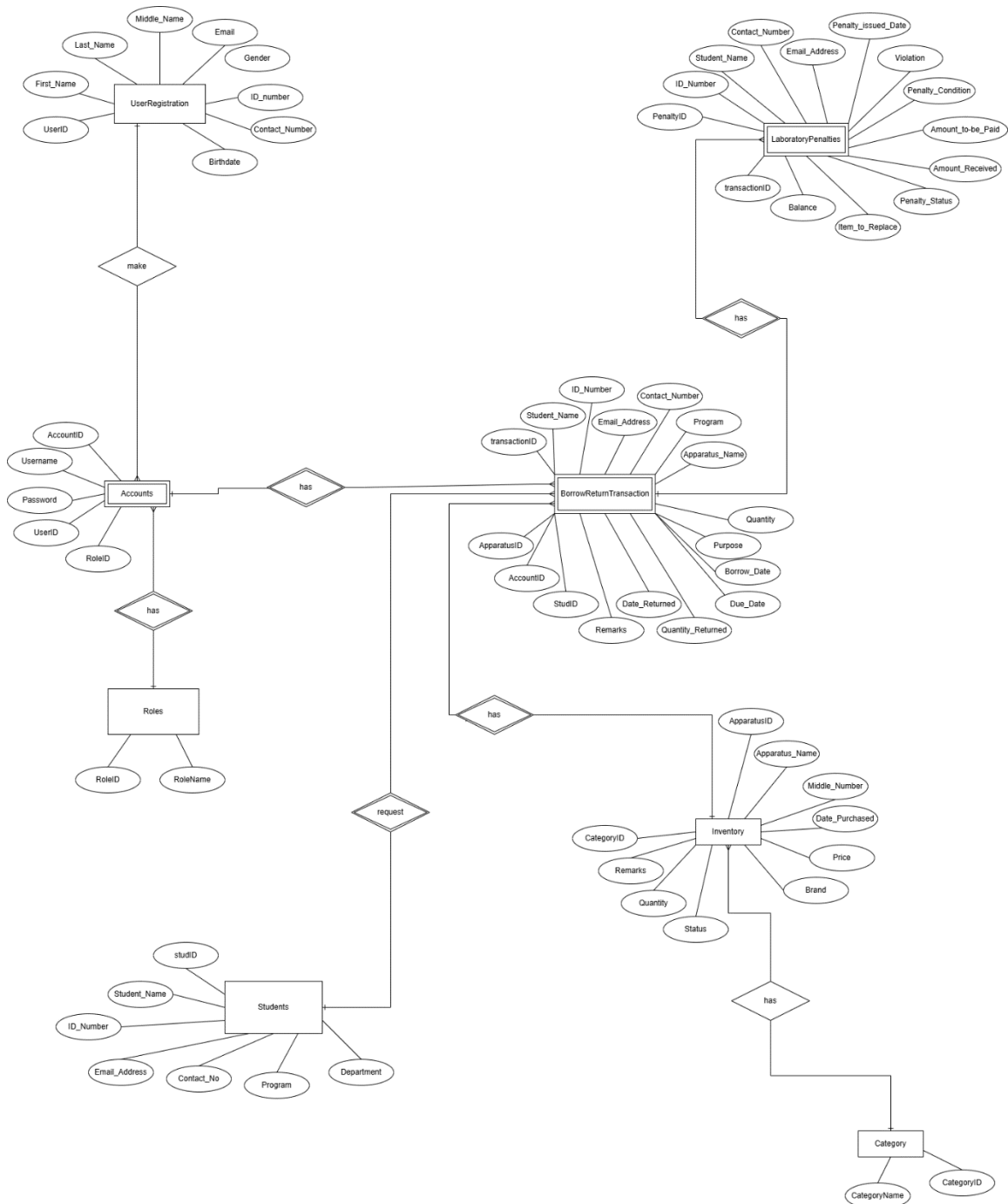


Figure 24. Entity Relationship Diagram

g. Database Schema

The following figure shows the Database Schema Diagram, which is used to represent the relationship between entities that makes up the database.

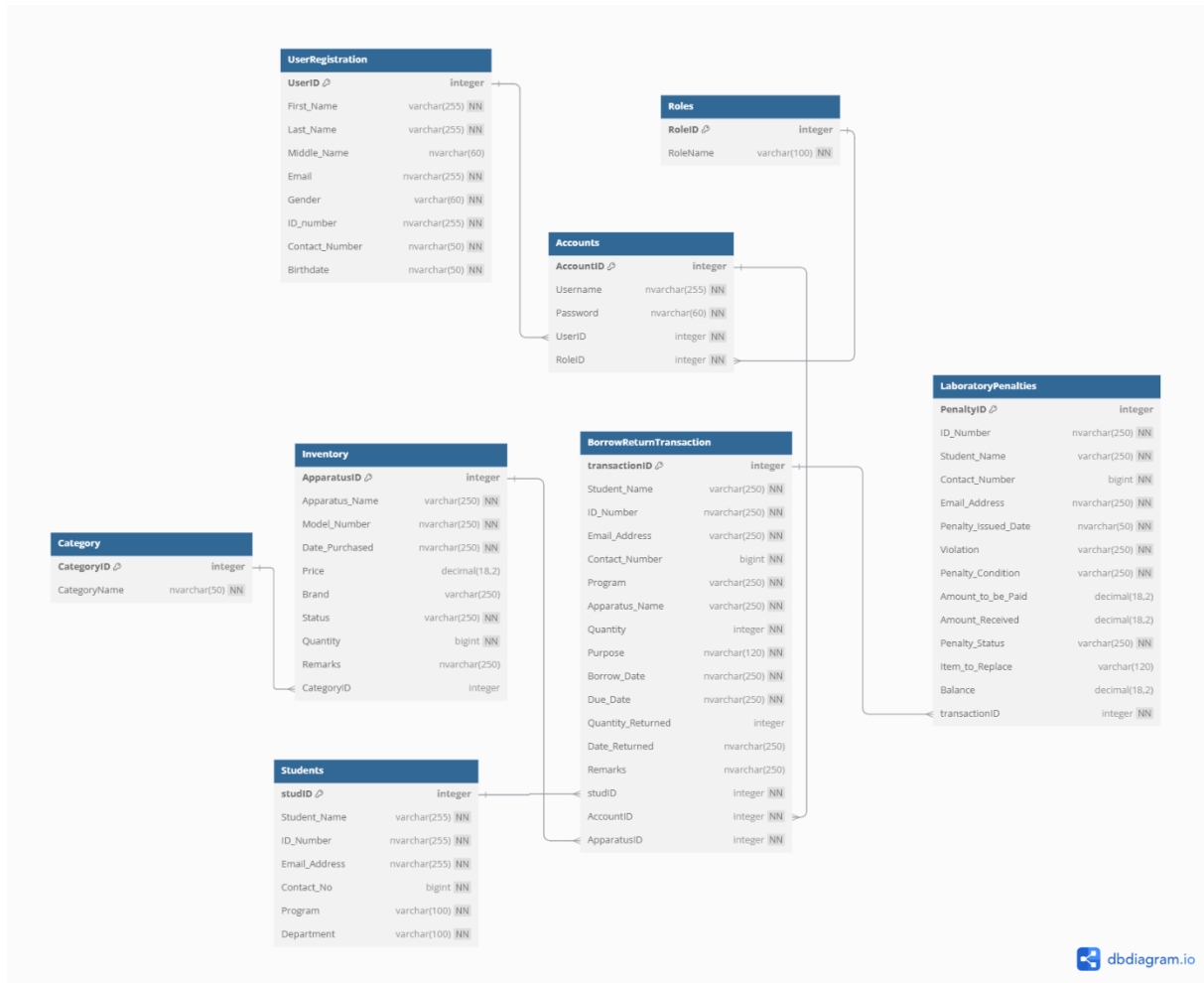


Figure 25. Database Schema

h. Implementation Strategies

The implementation strategies for modernizing the laboratory management system at the College of Engineering, Benedicto College, follow a structured and methodical approach to ensure a smooth transition from development to operational use. The process begins with a detailed project timeline that outlines each phase of development, testing, and deployment, ensuring that milestones are met on schedule. Resource allocation plays a crucial role in this phase, with dedicated teams assigned to various tasks, such as software development, testing, user training, and technical support. This allocation ensures that adequate personnel and tools are available for every critical step of the project.

During the deployment phase, risk management is addressed proactively. Potential risks, such as software bugs, system downtime, or user resistance, are identified early on, and contingency plans are developed to mitigate these risks. This includes creating backups, establishing troubleshooting protocols, and maintaining open communication with stakeholders to promptly resolve issues as they arise.

Additionally, quality assurance is a core aspect of the implementation strategy. The system undergoes multiple rounds of testing, from unit tests to full-scale system test, to ensure all functionalities perform as expected. Continuous feedback loops with selected testers allow for iterative improvements, ensuring the system meets user expectations.

Finally, the system rollout is done gradually, with user training sessions scheduled to introduce the laboratory staff to the new desktop application. Comprehensive user guides and documentation are provided to assist in the learning process. This structured approach ensures a smooth deployment of the system, transforming laboratory management processes into more efficient, reliable, and user friendly operations.

i. System User Interface

The System User Interface (System UI) is the collection of visual elements provided by your operating system, like the status bar, navigation controls, and notification panel, that let you interact with and control your device. It's the fundamental graphical layer you use to manage core device functions and settings.

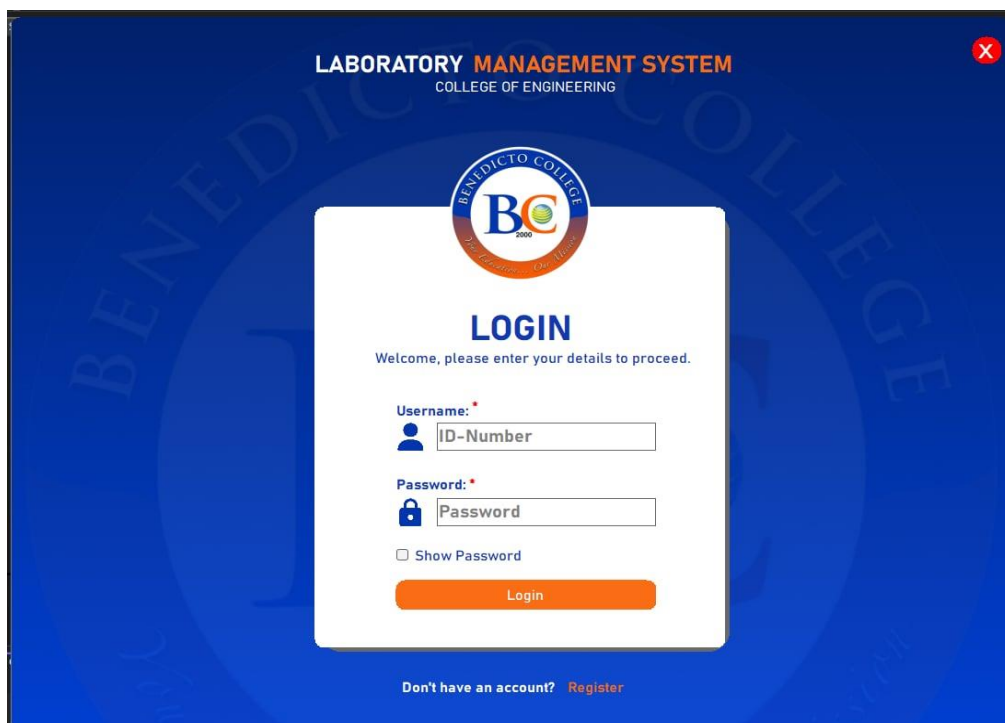
The image shows a login form for a 'LABORATORY MANAGEMENT SYSTEM' at 'COLLEGE OF ENGINEERING'. The background is dark blue with a large, faint watermark of 'BENEDICTO COLLEGE'. At the top center, the system name is in orange and white. Below it is the Benedicto College logo, which is a circular emblem with 'BC' and '2000' inside. The form itself is a white rounded rectangle. It has a 'LOGIN' heading in blue, followed by a welcome message. There are two input fields: 'Username:' with a red asterisk and a user icon, and 'Password:' with a red asterisk and a lock icon. The password field has a 'Show Password' checkbox below it. An orange 'Login' button is at the bottom of the form. At the very bottom of the form, there is a link: 'Don't have an account? Register'.

Figure 26: Login Form

Description: This is the entry point for all users. It includes fields for a username (which is the ID number) and password. There's an option to show the password for convenience, and a button for users to register if they don't have an account.

The screenshot displays the registration interface for the Benedicto College Laboratory Inventory System. At the top, the Benedicto College logo is centered above the text "LABORATORY INVENTORY SYSTEM" and "COLLEGE OF ENGINEERING". The main form is titled "REGISTRATION" and includes the instruction "Provide your personal information to register." The form is divided into two main sections: a registration form on the left and a login details section on the right.

Registration Form Fields:

- ID Number:** A text input field with a red asterisk. Below it, a note states "Your ID Number will be your username:".
- Set Role:** A dropdown menu with a red asterisk.
- First Name:** A text input field with a red asterisk.
- Middle Name:** A text input field with a red asterisk.
- Last Name:** A text input field with a red asterisk.
- Date of Birth:** A date picker field with a red asterisk.
- Gender:** Radio buttons for "Male" and "Female" with a red asterisk.
- Email:** A text input field with a red asterisk.
- Contact No:** A text input field with a red asterisk.

Login Details Section:

- LOGIN DETAILS** (Section Header)
- Password:** A text input field with a red asterisk. A note below it states "Password must be at least 8 characters, including letters and numbers.".
- Confirm Password:** A text input field with a red asterisk.
- Show Password:** A checkbox.
- Register:** An orange button.
- Clear all:** A link.
- Back to Login:** A link.

Figure 27: Registration Form

Description: When users choose to register, they must fill in details such as their ID number(which will serve as their username), role (student, staff, etc.), and password with aconfirmation. This registration requires admin approval before the account becomes active, ensuring controlled access to the system.

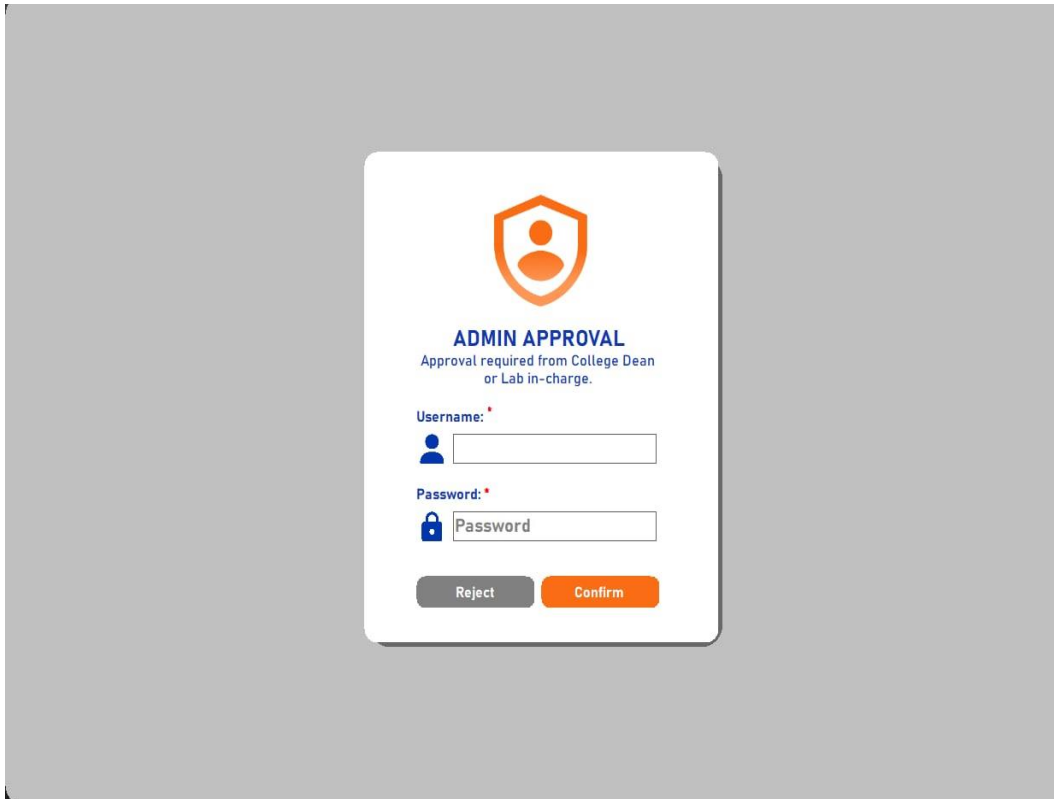
The image shows a central white rectangular dialog box with rounded corners, set against a solid grey background. At the top of the dialog is an orange shield icon containing a stylized human figure. Below the icon, the text 'ADMIN APPROVAL' is displayed in a bold, blue, sans-serif font. Underneath this, in a smaller blue font, is the text 'Approval required from College Dean or Lab in-charge.' The form contains two input fields: the first is labeled 'Username:' with a small red asterisk and a blue person icon to its left; the second is labeled 'Password:' with a small red asterisk and a blue padlock icon to its left. The password field has the word 'Password' written inside it. At the bottom of the dialog are two buttons: a grey button labeled 'Reject' and an orange button labeled 'Confirm'.

Figure 28: Admin Approval

Description: The admin approval interface with a shield icon. It requires the user to enter their username and password for authentication. The user then has the option to reject or confirm the approval request.

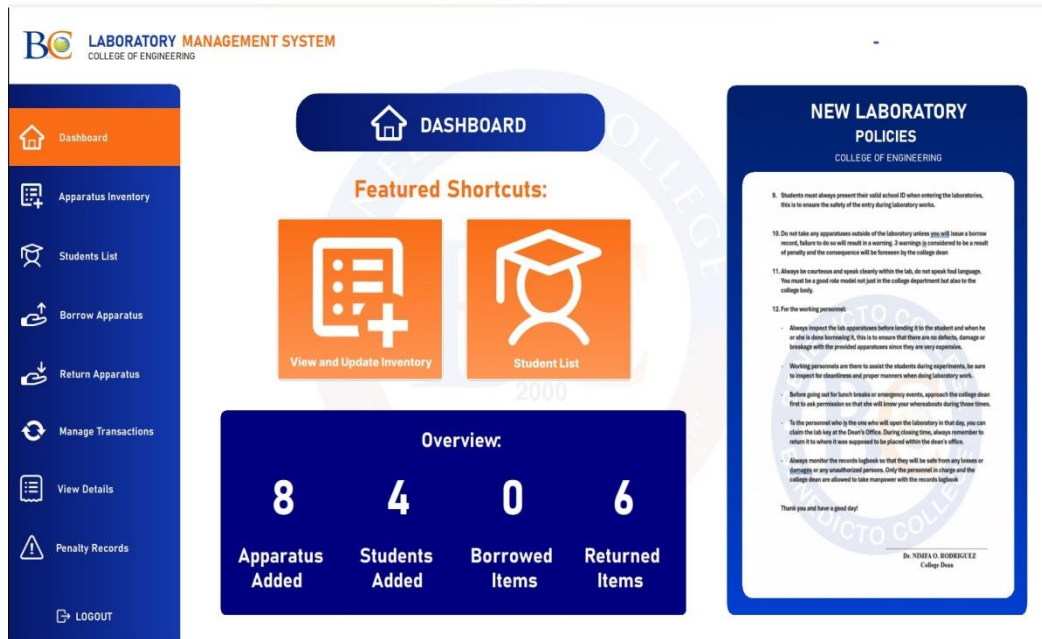


Figure 29: Dashboard

Description: This is the central hub for users after logging in. It provides an overview of the number of apparatus added, students registered, items borrowed, and items returned. There's also a section for new laboratory policies, along with shortcut buttons that lead to different management sections such as the inventory, student list, and borrowing/return processes. Navigation tabs include apparatus inventory, student list, borrowing and returning apparatus, transaction management, reports, and penalty records.

Figure 30: Add Apparatus

Description: This feature allows the admin to input new laboratory apparatus into the system, including details such as the item name, description, quantity, and any other necessary data. Once added, these items will be available for students to borrow.

Figure 31: Add Student Record

Description: This function is used to add students to the system. The admin can input student details, allowing them to borrow apparatus. This ensures that only registered students are allowed to make transactions in the system.

The screenshot displays the 'Borrowing Apparatus' form within the Laboratory Management System. The interface includes a sidebar with navigation options: Dashboard, Apparatus Inventory, Students List, Borrow Apparatus (highlighted), Return Apparatus, Manage Transactions, View Details, Penalty Records, and LOGOUT. The main form area is titled 'Borrowing Apparatus' and features a 'NOTE: The student can only borrow up to 10 equipments.' Below this, there is a section for 'Apparatus to borrowed:' with five rows, each containing a dropdown menu for selecting an apparatus and a 'Quantity' input field. To the left of this section, there is a 'Enter student's ID No:' field with a 'Search' button. Below the search field, there are input fields for 'Student Name:', 'ID Number:', 'Email Address:', 'Contact Number:', and 'Program:'. At the bottom left of the form, there is a 'Clear all' button. On the right side of the form, there are two date pickers: 'Borrowing Date:' (set to 10/21/2024 10:28 pm) and 'Due Date:' (set to 10/12/2024 08:58 pm). A 'Borrowing Purpose:' text area is also present. A green 'Confirm' button is located at the bottom right of the form.

Figure 32: Apparatus Borrowing

Description: This section is where the admin or laboratory personnel records the details of apparatus that students want to borrow. The information includes which apparatus is being borrowed, the borrowing student's ID, and the duration of the loan. This creates a formal record for tracking purposes.

Description: The inventory list displays all available laboratory apparatus. The admin can update the details of each apparatus, add new items, or remove outdated or damaged equipment. This ensures the inventory is always up-to-date for students to borrow.

LABORATORY MANAGEMENT SYSTEM
COLLEGE OF ENGINEERING

Update Borrow and Return Transactions
Manage Transactions

Search: Refresh

Student Name	ID Number	Email Address	Contact Number	Program	Apparatus Name	Quantity	Purpose	Borrow Date	Due Date	Quantity Returned	Date Returned	Remarks
Charlie Mike Romo	21-2331081	charlieP@gmail.com	9956629081	BSME	Hot Plates	2	Do Activity	Oct 12 2024 9:3	Oct 14 2024 10:3	2	Oct 13 2024 11:4	Item has been ret.
Charlie Mike Romo	21-2331081	charlieP@gmail.com	9956629081	BSME	Safety Goggles	1	Do Activity	Oct 12 2024 9:3	Oct 14 2024 10:3	1	Oct 12 2024 11:3	Thank you
Charlie Mike Romo	21-2331081	charlieP@gmail.com	9956629081	BSME	Safety Goggles	1	To do some sury	Oct 13 2024 1:4	Oct 16 2024 8:6		Oct 21 2024 9:2	
Whomee Reddusau	21-2001683	whomeer552@gmail.com	9956629081	BSME	Pinless Screwdriver	1	On a some ches	Oct 18 2024 8:4	Oct 20 2024 8:8	1	Oct 19 2024 11:5	Item was returned
Thouze Cortes	21-2001976	louiseh@comcast.net	9473720086	BSCE	Head Hat	1	for class a surm	Oct 18 2024 11:2	Oct 19 2024 8:5	1	Oct 19 2024 3:5	Apparatus is still i
Thouze Cortes	21-2001976	louiseh@comcast.net	9473720086	BSCE	Safety Goggles	1	for class a surm	Oct 18 2024 11:2	Oct 19 2024 8:5	1	Oct 19 2024 12:0	This is a late retur

UPDATE BORROW AND RETURN INFORMATION

Clear all

ID Number:

Student Name:

Email Address:

Contact No:

Program:

Apparatus Name:

Borrowed Quantity:

Returned Quantity:

Borrow Date:

Due Date:

Date Returned:

Purpose:

Remarks:

Delete Export to Excel Update

Figure: 35: Manage Transactions

Description: Management Transactions allow users to update borrow and return transactions by searching for specific records, modifying relevant fields, and saving changes. Additional features include viewing transaction details, deleting transactions, and exporting data to Excel. The sidebar menu provides access to other system functionalities such as managing apparatus inventory, student lists, and penalty records.

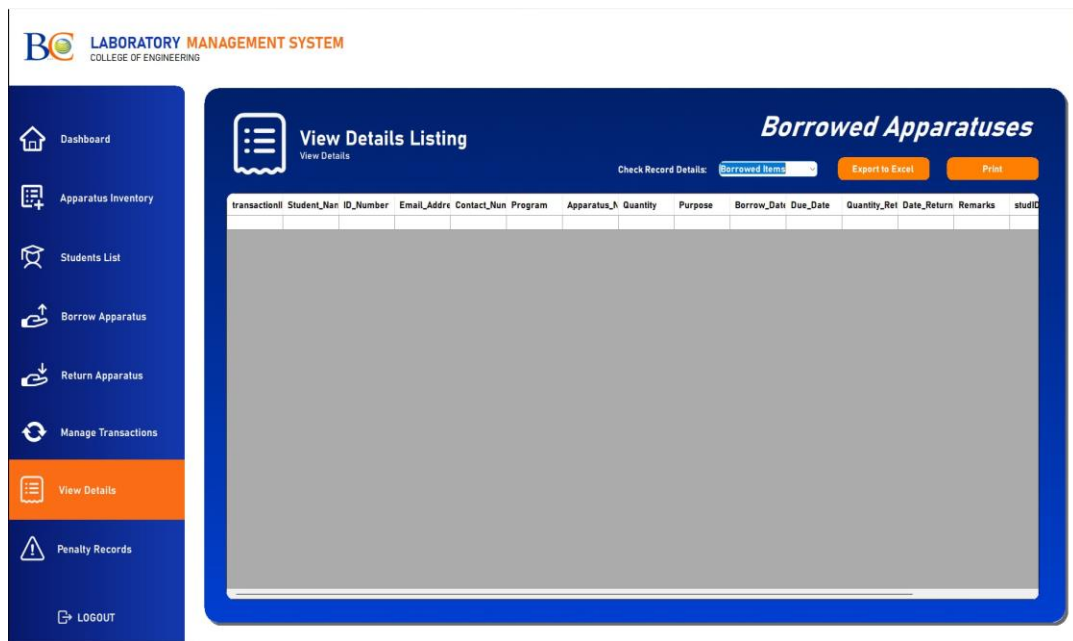


Figure 36: View Details Listing

Description: This section provides a complete log of all borrowing and returning transactions. Admins can view transaction histories, generate reports, export data to Excel, and print records. This allows for easy monitoring and auditing of all activities within the system.

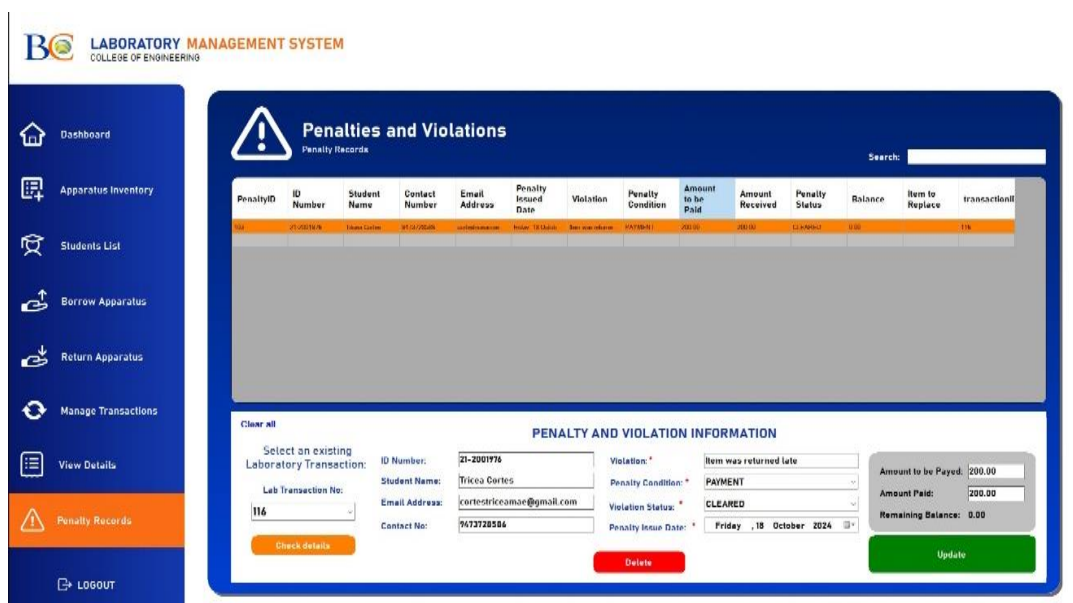


Figure 37: Penalties and Violations

Description: This section records penalties imposed on students who fail to return apparatus on time or who return damaged items. The penalty records ensure accountability and can include fines or restrictions on future borrowing.

LABORATORY MANAGEMENT SYSTEM
COLLEGE OF ENGINEERING

View Students Information
Students List - View Student Information

Filter by Program:

studID	Student_Name	ID_Number	Email_Address	Contact_No	Program	Department
1	Venus Angels	21-2012873	venus2@gmail.com	9357664512	BSIE	College Of Engin...
2	Charles Mike Thomas	21-2012881	charlie.thomas@gmail.com	9950102611	BSIE	College Of Engin...
3	Vincent Rodriguez	21-2012427	vincent402@gmail.com	9950105178	BSIE	College Of Engin...
4	Thomas Gomez	21-2012876	tdhobbes@gmail.com	9472722086	BSIE	College Of Engin...

Search for :

Enter student's ID No:

VIEW STUDENTS INFORMATION

Student Name: Contact No:

ID Number: Program:

Email Address: Department:

Figure 38: View Student Information

Description: This feature stores the details of all registered students, including their ID numbers, contact information, borrowing history, and any penalties they have incurred. It helps the admin manage student access and review individual borrowing patterns.

CHAPTER IV

Result and Analysis

The System Usability Scale (SUS) is a widely used tool to check how easy and user-friendly software, websites, and other technological systems are. It helps collect feedback from users and measure their thoughts about how easy it is to use a system. In our study, we used the SUS to see what participants thought about the usability of a specific system. This report shares the results, analysis, and discussion of the SUS survey. It gives us a better understanding of what participants experienced and what they think about how easy the system to use.

In this study, the researcher utilized the System Usability Scale (SUS) survey to assess how easy and practical the application was to use. The SUS survey consists of 10 questions, which were designed to gather feedback from users by presenting statements in both positive and negative ways. A total of 10 individuals participated in the survey, providing their valuable insights based on their experience with the application. Each question in the survey was rated on a scale of 1 to 5, allowing participants to indicate their level of agreement or disagreement.

1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

Table 2: 5-Point Likert Agreement Scale

1. I think that I would like to use this product frequently.

Scale:

24 responses

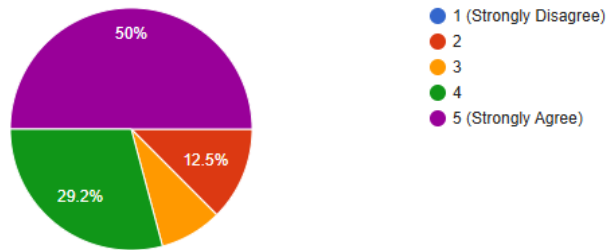


Figure 39: Analysis of User Acceptance Survey for Laboratory Management System

The user acceptance survey for the laboratory management system revealed positive overall feedback from respondents. Notably, 50% of participants expressed strong agreement with the statement "I think that I would like to use this product frequently." This indicates a high level of satisfaction and intention for continued use. 29.2% of respondents indicated a neutral response, suggesting a moderate level of acceptance. Only 12.5% disagreed with the statement, underscoring the general positivity surrounding the system. These results suggest that the system is well-received by users and aligns with their needs and preferences.

Further analysis of the survey data can provide deeper insights into specific aspects of user experience, such as ease of use, functionality, and perceived benefits. By identifying strengths and areas for improvement, the laboratory can make informed decisions to enhance the system's effectiveness and user satisfaction.

2. I found the product unnecessarily complex.

Scale:

24 responses

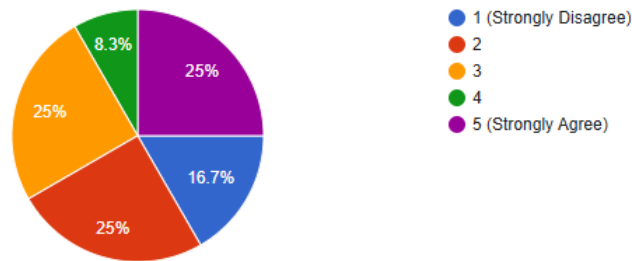


Figure 40: Analysis of User Perception of System Complexity

The second statement from the user acceptance survey, "I found the product unnecessarily complex," provides insights into the perceived ease of use of the laboratory management system. 25% of respondents agreed or strongly agreed with this statement, indicating that a significant portion of users found the system to be overly complex. Another 25% disagreed, suggesting that a portion of users found the system to be intuitive and straightforward. 16.7% expressed a neutral opinion, indicating that the system's complexity was neither strongly agreed or disagreed with.

These results highlight a need for further investigation into the specific areas of the system that users find complex. By identifying these challenges, the laboratory can take steps to simplify the user interface and improve the overall user experience. This could involve streamlining workflows, providing clearer instructions, or offering additional training resources.

3. I thought the product was easy to use.

Scale:

24 responses

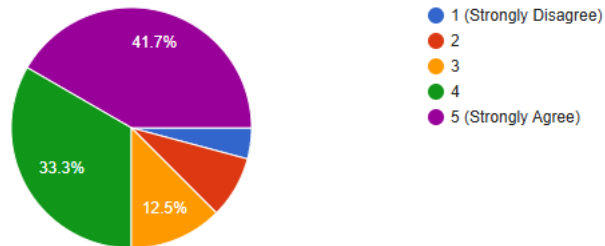


Figure 41: Analysis of User Perception of Ease of Use

The third statement from the user acceptance survey, "I thought the product was easy to use," directly assesses the perceived ease of use of the laboratory management system. 41.7% of respondents strongly agreed or agreed with this statement, indicating that a majority of users found the system to be intuitive and user-friendly. 33.3% expressed a neutral opinion, suggesting that the system's ease of use was neither strongly agreed or disagreed with. 12.5% disagreed with the statement, indicating that a smaller portion of users found the system to be difficult to navigate or understand.

These results align with the previous analysis of system complexity. The higher percentage of positive responses to this statement suggests that while some users may have found certain aspects of the system complex, the overall ease of use was generally perceived positively. However, the 12.5% of negative responses highlight the need for continued efforts to simplify the system and address any remaining usability challenges.

4. I think that I would need the support of a technical person to be able to use this product.

Scale:

24 responses

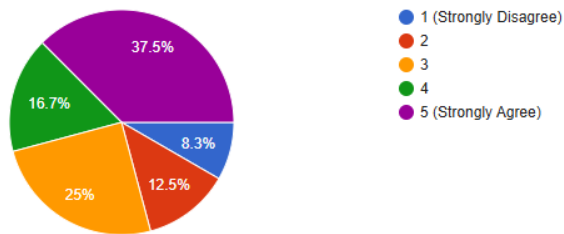


Figure 42: Analysis of Perceived Need for Technical Support

The fourth statement from the user acceptance survey, "I think that I would need the support of a technical person to be able to use this product," addresses the perceived level of technical support required for using the laboratory management system. 37.5% of respondents disagreed or strongly disagreed with this statement, indicating that a majority of users felt confident in using the system without extensive technical assistance. 25% expressed a neutral opinion, suggesting that while they may have required some support, they did not feel completely reliant on technical personnel. 16.7% agreed with the statement, indicating a need for moderate technical assistance, and 12.5% strongly agreed, suggesting a high level of reliance on technical support.

These results align with the previous analysis of system complexity and ease of use. The lower percentage of positive responses to this statement suggests that while the majority of users found the system to be relatively user-friendly, some still felt the need for additional support. This information can be used to inform the development of training materials, documentation, and technical support resources to address the specific needs of users.

5. I found the various functions in the product were well integrated.

Scale:

24 responses

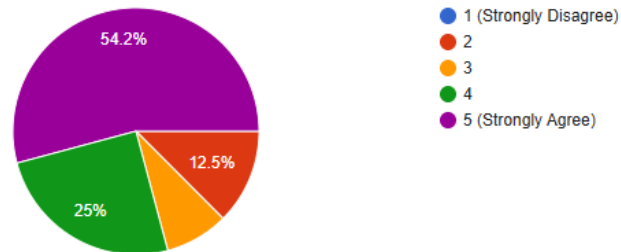


Figure 43: Analysis of Perceived Integration of Functions

The fifth statement from the user acceptance survey, "I found the various functions in the product were well integrated," assesses the perceived level of integration among the different functionalities of the laboratory management system. 54.2% of respondents strongly agreed or agreed with this statement, indicating that a majority of users found the system's functions to be well-connected and easy to navigate. 25% expressed a neutral opinion, suggesting that while the integration was generally good, there were some areas for improvement. 12.5% disagreed with the statement, indicating that a smaller portion of users found the functions to be disjointed or difficult to use together.

These results highlight the successful integration of the system's features, which is likely to contribute to a positive user experience. However, the 12.5% of negative responses suggest that there are still opportunities to further enhance the integration of certain functions, potentially by streamlining workflows or providing more intuitive navigation options.

6. I thought there was too much inconsistency in this product.

Scale:

24 responses

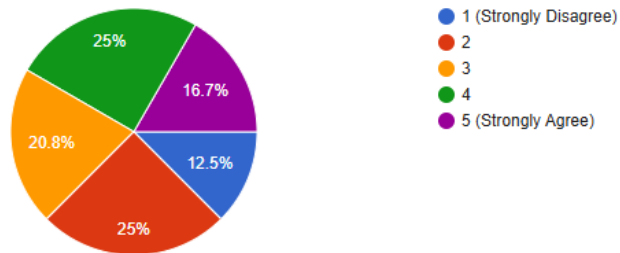


Figure 44: Analysis of Perceived Inconsistency in the Product

The sixth statement from the user acceptance survey, "I thought there was too much inconsistency in this product," addresses the perceived level of consistency within the laboratory management system. 25% of respondents disagreed or strongly disagreed with this statement, indicating that a significant portion of users found the system to be consistent and reliable. 25% agreed with the statement, suggesting that a notable portion of users perceived inconsistencies in the product. 20.8% expressed a neutral opinion, indicating that the system's consistency was neither strongly agreed or disagreed with. 12.5% strongly agreed with the statement, highlighting a significant concern about inconsistencies.

These results indicate that while a majority of users found the system to be consistent, there is a notable portion of users who experienced inconsistencies. Identifying the specific areas of inconsistency is crucial for improving the overall user experience. This could involve addressing issues with data integrity, workflow consistency, or user interface design.

7. I imagine that most people would learn to use this product very quickly.

Scale:

24 responses

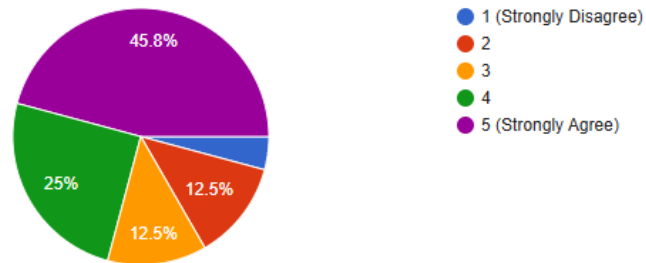


Figure 45: Analysis of Perceived Ease of Learning

The seventh statement from the user acceptance survey, "I imagine that most people would learn to use this product very quickly," assesses the perceived ease of learning for the laboratory management system. 45.8% of respondents strongly agreed or agreed with this statement, indicating that a majority of users believe that the system is intuitive and easy to learn. 25% expressed a neutral opinion, suggesting that while the system is generally easy to learn, there may be some areas that require more effort. 12.5% disagreed with the statement, indicating that a smaller portion of users believe that learning the system would be challenging.

These results align with the previous analysis of system complexity and ease of use. The high percentage of positive responses to this statement suggests that the system's design and interface contribute to a positive learning experience. However, the 12.5% of negative responses highlight the need for continued efforts to improve the system's onboarding process and provide adequate training resources to support users in learning the system effectively.

8. I found the product very awkward to use.

Scale:

24 responses

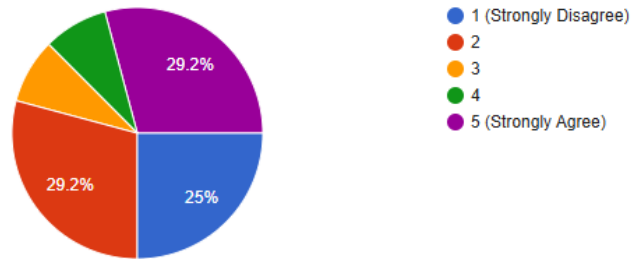


Figure 46: Analysis of Perceived Awkwardness in Product Use

The eighth statement from the user acceptance survey, "I found the product very awkward to use," directly assesses the perceived usability of the laboratory managementsystem. 29.2% of respondents disagreed or strongly disagreed with this statement, indicating that a significant portion of users found the system to be intuitive and user friendly. 29.2% agreed with the statement, suggesting that a notable portion of users experienced awkwardness or difficulty in using the system. 25% expressed a neutral opinion, indicating that the system's usability was neither strongly agreed or disagreed with.

These results highlight the need for further investigation into the specific areas of the system that users find awkward. By identifying these challenges, the laboratory can take steps to improve the user interface, streamline workflows, or provide more intuitivenavigation options. This will contribute to a more positive and efficient user experience.

9. I felt very confident using the product.

Scale:

24 responses

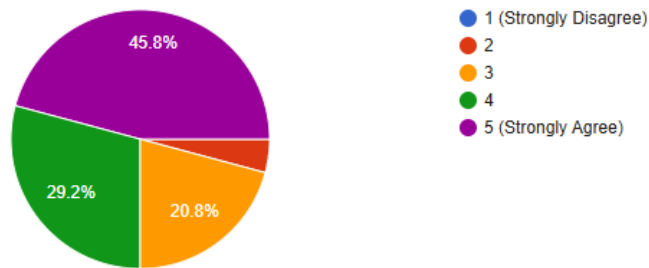


Figure 47: Analysis of Perceived Confidence in Using the Product

The ninth statement from the user acceptance survey, "I felt very confident using the product," assesses the overall level of user confidence in the laboratory management system. 45.8% of respondents strongly agreed or agreed with this statement, indicating that a majority of users felt confident and comfortable using the system. 29.2% expressed a neutral opinion, suggesting that while they were generally confident, there were some areas where they felt less certain. 20.0% disagreed with the statement, indicating that a smaller portion of users felt unsure or uncomfortable using the system.

These results align with the previous analysis of system complexity, ease of use, and perceived technical support. The high percentage of positive responses to this statement suggests that the system's design, features, and training materials contribute to a positive user experience and boost user confidence. However, the 20.0% of negative responses highlight the need for continued efforts to address any remaining challenges that may be affecting user confidence, such as improving system responsiveness or providing more targeted training resources.

10. I needed to learn a lot of things before I could get going with this product.

Scale:

24 responses

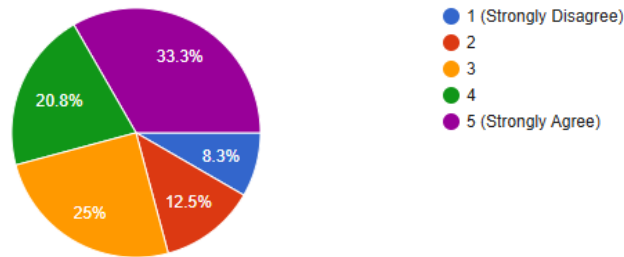


Figure 48: Analysis of Perceived Learning Curve

The tenth and final statement from the user acceptance survey, "I needed to learn a lot of things before I could get going with this product," assesses the perceived learning curve associated with using the laboratory management system. 33.3% of respondents disagreed or strongly disagreed with this statement, indicating that a significant portion of users found the system relatively easy to learn. 25% agreed with the statement, suggesting that a notable portion of users felt that a significant learning curve was involved. 20.8% expressed a neutral opinion, indicating that the learning curve was neither strongly agreed or disagreed with. 12.5% strongly agreed with the statement, highlighting a significant challenge in learning the system.

These results align with the previous analysis of perceived ease of learning and perceived need for technical support. While a majority of users found the system to be relatively easy to learn, there is still a portion of users who experienced a steeper learning curve. This information can be used to inform the development of more effective training materials, documentation, and onboarding processes to address the specific needs of users and reduce the perceived learning curve.

Overall, the comments from the System Usability Scale form, provide valuable feedback that can be used to further refine and enhance the laboratory management system focused on the system's ease of use, functionality, and integration. Users appreciated the intuitive interface,

streamlined workflows, and well-integrated features. Some comments specifically highlighted the system's ability to improve efficiency and accuracy in laboratory operations.

To determine the system's usability score, each respondent's SUS score was calculated using a specific formula: $SUS\ Score = (X + Y) \times 2.5$. Here, X represents the sum of points for all odd-numbered questions minus 5, and Y is calculated as 25 minus the sum of points for all even-numbered questions.

The SUS score serves as an indicator of the application's performance in terms of usability, encompassing factors like effectiveness, efficiency, and overall ease of use.

Below is a general guideline to help interpret the SUS score:

SUS Score	Grade	Adjective Rating
>80.3	A	Excellent
68-80.3	B	Good
68	C	Okay
51-68	D	Poor
<51	E	Awful

Table 3: SUS Score Guidelines

SUS Score	Grade	Adjective Rating
73.75	B	Good

Table 4: Overall SUS Score with Grade and Adjective Rating

By deploying the formula for the SUS Score, the score for the System Usability Scale was obtained to be 73.75. In his review, UIUX Trend website stated that this score was classified under "Good" for usability purposes. On the balances, the information of the data shows positive evaluation of the application usability.

Respondent No.	SUS Score	Grade	Adjective Rating
1	90	A	Excellent
2	87.5	A	Excellent
3	85	A	Excellent
4	87.5	A	Excellent
5	90	A	Excellent
6	90	A	Excellent
7	50	E	Awful
8	50	E	Awful
9	50	E	Awful
10	57.5	D	Poor

Table 5: Respondent's Individual SUS Score with Grade and Adjective Rating

CHAPTER V

Discussion of Results

The user acceptance survey for the laboratory management system provided valuable insights into user perceptions and experiences with the system. The results largely aligned with the study's objectives, demonstrating positive overall acceptance and satisfaction among users. The high levels of agreement with statements regarding ease of use, functionality, and integration indicate that the system effectively meets the needs of its intended users.

However, the survey also revealed areas for improvement. While the majority of users found the system to be intuitive and user-friendly, a portion of users expressed concerns about system complexity, inconsistencies, and the need for additional technical support. These findings highlight the importance of ongoing efforts to refine the system's design, address usability challenges, and provide comprehensive training and support resources.

The results of this study have practical implications for the laboratory. By addressing the identified areas for improvement, the laboratory can enhance the system's effectiveness, improve user satisfaction, and optimize laboratory operations. Furthermore, the findings can inform future system updates and developments, ensuring that the system continues to meet the evolving needs of its users.

It is important to note that this study has limitations. The sample size may not be representative of all potential users, and the survey may not capture the full range of user experiences. Future research could involve a larger sample size, longitudinal studies to track user satisfaction over time, and qualitative methods to gain deeper insights into user perspectives.

Overall, the user acceptance survey provides valuable insights into the laboratory management system's effectiveness and areas for improvement. By addressing the identified

challenges and leveraging the positive feedback, the laboratory can continue to optimize the system and ensure its long-term success.

CHAPTER VI

Conclusion and Recommendations

A. Conclusion

The research successfully developed and evaluated a new and modernized laboratory management system for the College of Engineering at Benedicto's College. The system effectively addresses the inefficiencies and concerns of the traditional paper-based system, providing a well-structured platform for managing laboratory records, minimizing manual processes, improving security and privacy, and implementing a built-in search functionality.

The system's key features, including modules for user registration, security, student records, apparatus management, borrowing and returning records, and penalty records, have significantly enhanced the efficiency and accuracy of laboratory operations. By automating tasks and providing a centralized platform for managing information, the system has reduced the time and effort required for administrative tasks, allowing laboratory personnel to focus on their core responsibilities.

While the system has demonstrated its effectiveness in addressing the research objectives, there are areas for further improvement. The lack of live chat support, direct contact with the property custodian, and reservation records could be considered limitations of the current system. Future research could explore the integration of these features to enhance the system's functionality and user experience.

Overall, the development and implementation of this modernized laboratory management system represent a significant advancement for the College of Engineering. By addressing the challenges of the traditional system and providing a more efficient and secure solution, the system has the potential to improve the overall performance of the engineering department.

B. Recommendations

Based on the research conducted, several practical suggestions can be made to enhance the effectiveness and usability of the laboratory management system. These include expanding the system's functionality to include features like live chat support, direct contact with the property custodian, and reservation records; implementing additional security measures; refining the search function for improved efficiency; and optimizing the user interface for better usability. Additionally, providing comprehensive training and ongoing support can help users effectively utilize the system. Future research could explore the system's scalability, long-term impact, and interoperability with other campus systems. Furthermore, the researchers recommend implementing the SMTP usage using the modern O2Auth method to enhance security and improve email communication. By implementing these suggestions, the laboratory management system can be further optimized to meet the needs of its users, improve efficiency, and contribute to the overall success of the College of Engineering.

REFERENCES

Websites:

- [1] eLabNext. (n.d.). <https://www.elabnext.com/>
- [2] UBC Sauder. (n.d.). CLC book and equipment borrowing. <https://mybcom.sauder.ubc.ca/clc/book-and-equipment-borrowing>
- [3] Quartzy. (n.d.). Quartzy. <https://www.quartzy.com/>
- [4] LabVantage. (n.d.). LabVantage. <https://www.labvantage.com/>
- [5] LabArchives. (n.d.). LabArchives. <https://www.labarchives.com/>

APPENDICES

Appendix A. System Specifications

Hardware Specification

System Manufacturer:	Dell Inc.
System Model:	Latitude E7450
Processor:	Intel(R) Core(TM) i5-5300U CPU @ 2.30GHz (4 CPUs), ~2.3GHz
Memory:	8192MB RAM
Hard disk:	128GB SSD
Mouse:	Dell Multi-device Wireless Mouse - MS5320W

Software Specification

Operating System: Windows 10 Pro

Additional libraries and software dependencies:

- SQL Server
- SQL Server Management Studio 19 (or latest)
- Visual Basic Redistributable 2009 C++ (optional framework)
- Visual Basic Redistributable 2015 .NET (main library)
- Zero One Tier local domain (for multiple user setup)

Appendix B. Flow of the System

Current Flow of the System

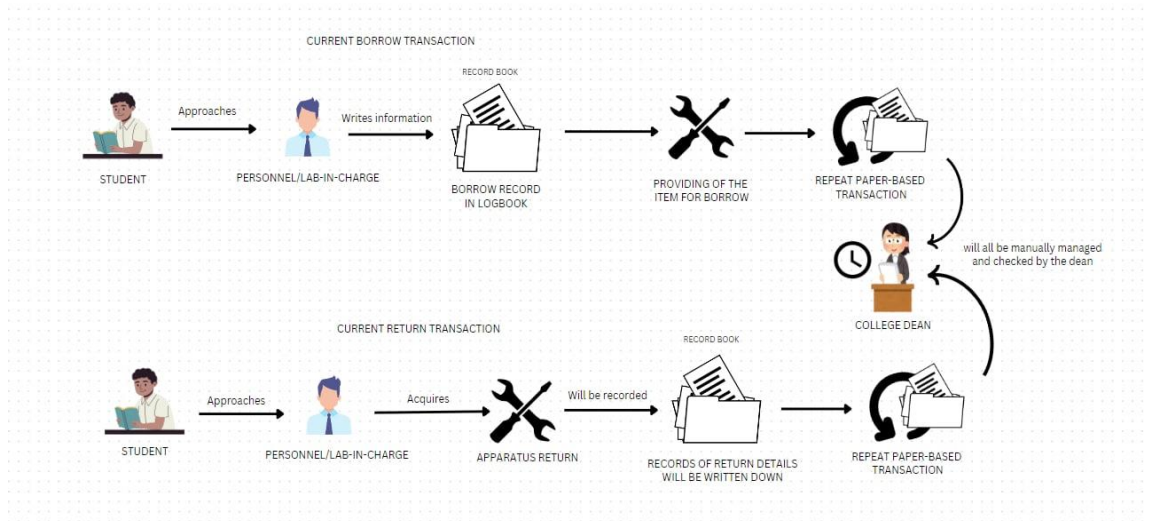


Figure 49: Current Flow of the Borrow Transaction

The current flow of the system for borrow transactions involves students approaching the Personnel Lab-in-Charge to borrow items. The Personnel writes the student's information in a record book and provides the item for borrowing. The records of return details are manually written down by the College Dean. This process is repeated for each borrow transaction, with no automated system in place.

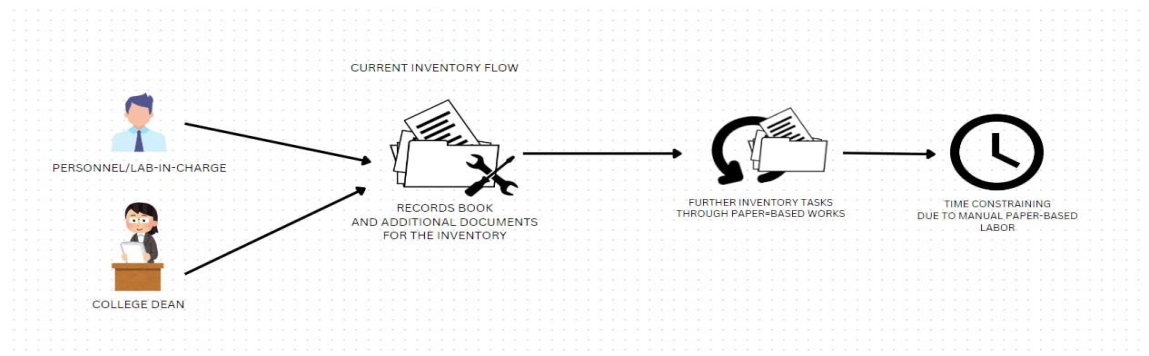


Figure 50: Current Inventory Flow

The current inventory flow involves the Personnel Lab-in-Charge, who maintains records and additional documents related to the inventory in a record book. Further inventory tasks are carried out manually using paper-based methods, which can be time-consuming due to the manual labor involved. The College Dean is involved in overseeing the inventory management process.

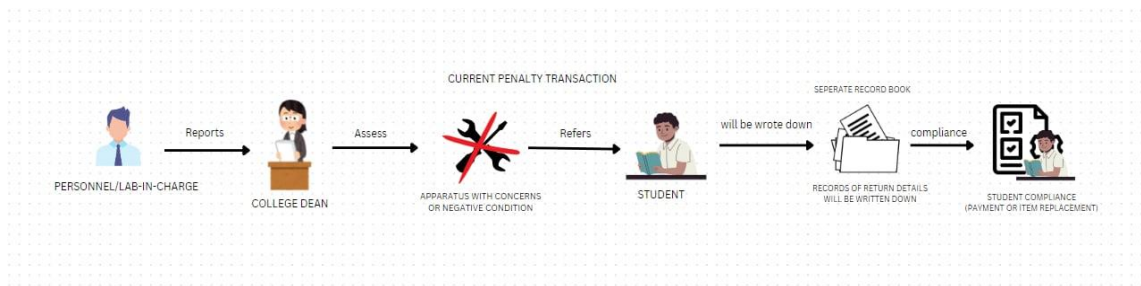


Figure 51:Current Flow of the Penalty Transaction

The current flow of the system for penalty transactions involves the College Dean, Personnel Lab-in-Charge, and the student. If apparatus is returned in a damaged or negative condition, the Personnel Lab-in-Charge assesses the issue, refers to the student's borrowing transaction, and records the details in a separate record book. The College Dean then implements the penalty, which may involve direct student compliance or payment.

Proposed Flow of the System

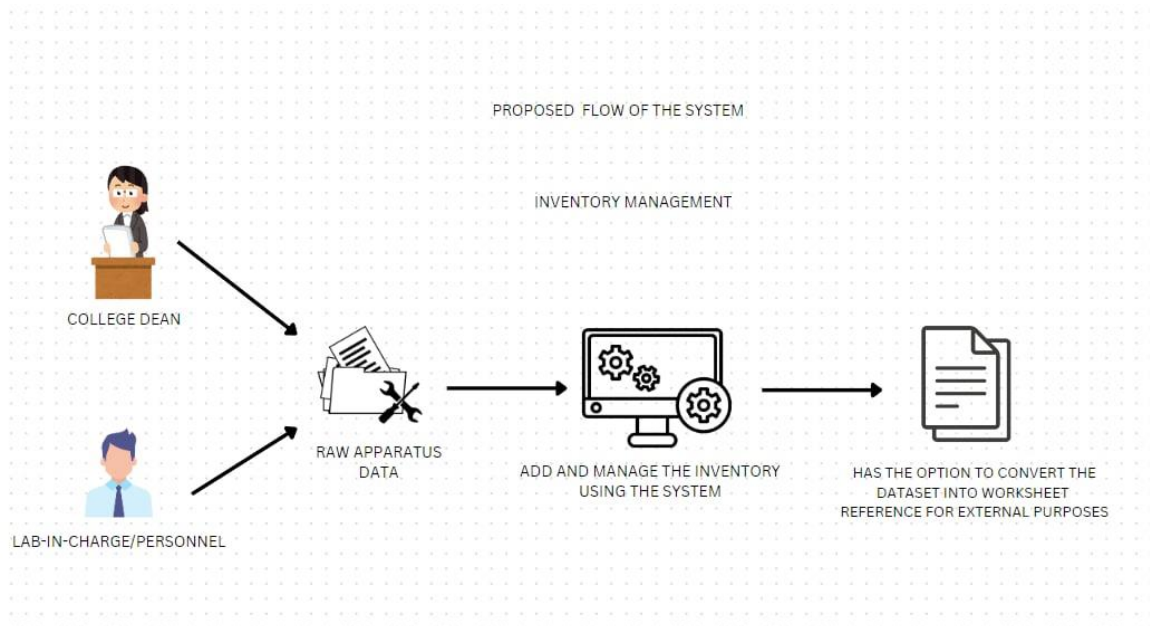


Figure 52: Proposed Flow of the Inventory Management

The proposed flow of the system for inventory management involves the College Dean and Lab-in-charge/Personnel. The Dean oversees the system and has access to raw apparatus data. Lab-in-charge/Personnel are responsible for adding and managing the inventory using the system. The system allows for the conversion of the dataset into a worksheet for external reference, providing flexibility for data analysis and reporting.

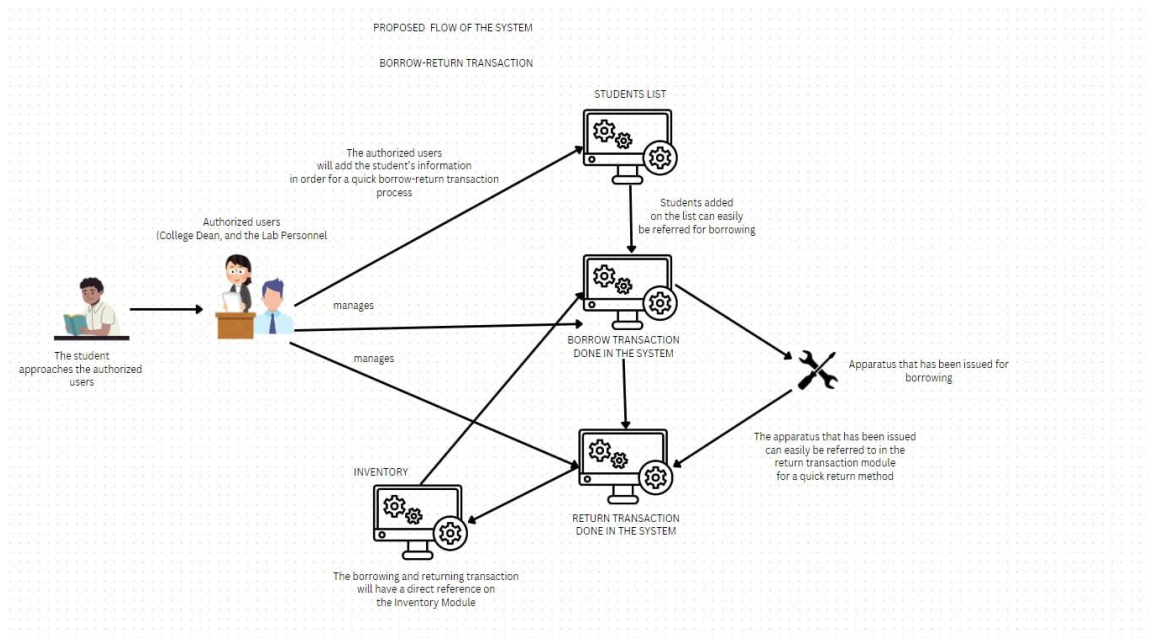


Figure 53: Proposed Flow of the Borrow-Return Transaction

The proposed flow of the system for borrow and return transactions involves the College Dean, Lab-in-charge/Personnel, and students. The Dean and Lab-in-charge/Personnel manage the system, including student information and inventory. Students can easily be added to the system for borrowing purposes. The borrowing and return transactions are done within the system, with the inventory module updated accordingly. The issued apparatus can be easily referenced for return transactions, optimizing the process. The authorized users who has access to the system has different user roles with functional hierarchy, where the user with an admin role can easily manage all of the system's records while the ones being marked as the normal user are limited in certain areas of the system.

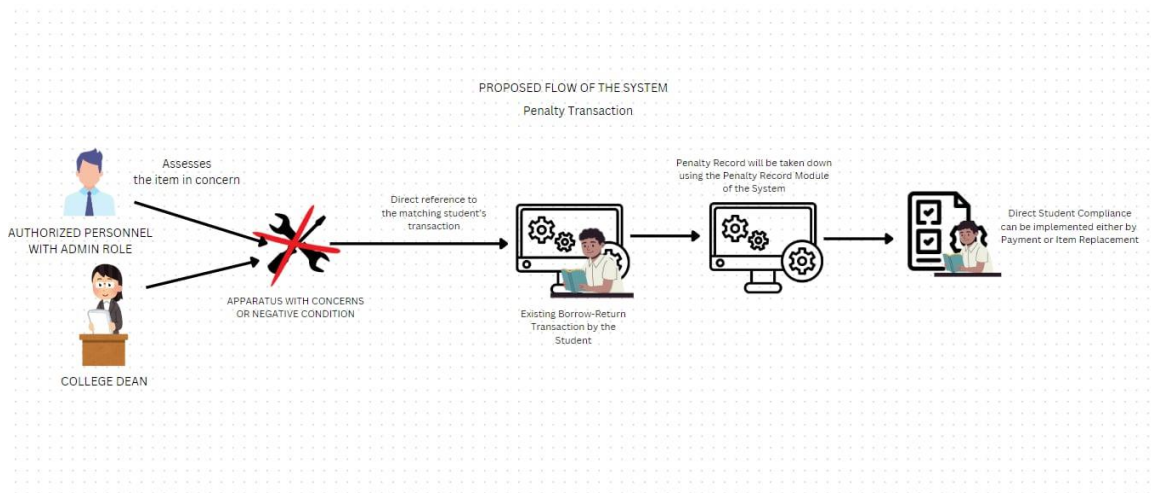
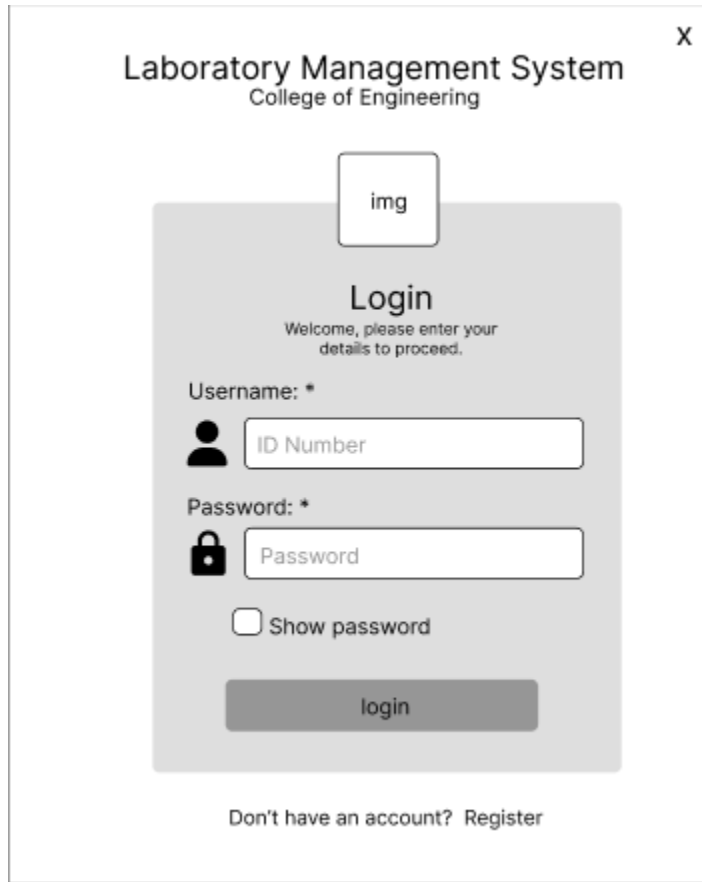


Figure 54: Proposed Flow of the Penalty Transaction

The proposed flow of the system for penalty transactions involves authorized personnel with an admin role. When apparatus is returned in a damaged or negative condition, the authorized personnel assesses the issue and refers to the student's borrowing transaction. A penalty record is then created using the Penalty Record Module of the system. The penalty can be implemented through direct student compliance or payment.

Appendix C. Application Prototype



The image shows a login form prototype for a 'Laboratory Management System' at the 'College of Engineering'. The form is contained within a light gray rectangular box with a close button 'X' in the top right corner. At the top of the form is a placeholder for a logo, labeled 'img'. Below this is the title 'Login' in a bold font, followed by a welcome message: 'Welcome, please enter your details to proceed.' The form contains two input fields: 'Username: *' with a user icon and a text box labeled 'ID Number', and 'Password: *' with a lock icon and a text box labeled 'Password'. Below the password field is a checkbox labeled 'Show password'. A dark gray 'login' button is positioned below the inputs. At the bottom of the form, there is a link: 'Don't have an account? Register'.

Laboratory Management System
College of Engineering

img

Login
Welcome, please enter your details to proceed.

Username: *
ID Number

Password: *
Password

☐ Show password

login

Don't have an account? Register

Figure 55: Login Form



Laboratory Management System

College of Engineering

REGISTRATION

Provide your personal information to register

ID Number: * Set role: *

Your ID Number will be your username

First Name: *

Middle name:

Last name: *

Date of birth: *

Gender: * ☐ Male ☐ Female

Email: *

Contact No : *

LOGIN DETAILS

Password must be at least 8 characters, including letters and numbers.

Password: *

Confirm: *

☐ Show password

[Back to login](#)

Figure 56:Registration Form



Laboratory Management System
College of Engineering

ID NUMBER - ROLE
Name

- Dashboard
- Apparatus Inventory
- Student List
- Borrow Apparatus
- Return Apparatus
- Manage Transactions
- View Details
- Penalty Records
- LOGOUT

DASHBOARD

Featured Shortcuts:



View and Update Inventory



Student List

Overview

0
0
0
0

Apparatus Added
Students Added
Borrowed Items
Returned Items

NEW LABORATORY POLICIES

College of Engineering



Figure 57: Dashboard

img

Apparatus inventory

Add an apparatus

Apparatus name: *

Stocks Quantity: *

Model number:

Category: *

Date purchased: *

Item remarks:

Price:

Brand:

Status: *

Save

Clear all

Figure 58: Add Apparatus

img

Student list

Add a student

Student name: *

ID Number: *

Contact No: *

Email: *

Program: *

Department: *

Add

Clear all

Figure 59: Add Student

X

img

ADMIN APPROVAL

Approval required from the College Dean
or Lab in-charge.

Username: *

ID Number

Password: *

Password

Reject

Confirm

Figure 60: Admin Approval


 Borrowing Apparatus

Enter student's ID No:

Student name:
 ID Number:
 Contact No:
 Email:
 Program: ▼

Borrowing Apparatus

NOTE: The student can only borrow up to 10 equipments

Apparatus to be borrowed:

1.		Quantity:		▲ ▼
	Available quantity:			
2.		Quantity:		▲ ▼
	Available quantity:			
3.		Quantity:		▲ ▼
	Available quantity:			
4.		Quantity:		▲ ▼
	Available quantity:			
5.		Quantity:		▲ ▼
	Available quantity:			

Borrowing date: *

Due date: *

Borrowing purpose: *

Figure 61: Borrowing Apparatus

img

Return Apparatus Transactions

Return Apparatus

img

Return Apparatus

Enter student's ID No:

search

RETURN APPARATUS INFORMATION

X

Apparatus name: *

Quantity: *

↑

↓

Borrowed date: *

Due date: *

Date returned: *

Item remarks:

Update

Clear all

Figure 62: Return Apparatus



Update Borrow and Return Transactions

Manage Transactions

Search:

Clear all

UPDATE BORROW & RETURN INFORMATION

ID number:

Student name:

Email:

Contact No:

Program:

Apparatus name: *

Borrowed date: *

Borrowed quantity: *

▲▼

Due date: *

Returned quantity: *

▲▼

Date returned: *

Purpose: *

Item remarks: *

Delete

Export to excel

Update

Figure 63: Borrow/Return Transactions

img

View and Update Inventory List

Apparatus Inventory

Filter by category: Filter by status:

Search:

Refresh

Clear all

X

VIEW AND UPDATE INVENTORY INFORMATION

Apparatus name: *

Brand:

Item remarks: *

Model number:

Status: *

Date purchased: *

Stocks Quantity: *

Price:

Category: *

Delete

Export to excel

Update

Figure 64: View and Update Inventory List

Penalties and Violations
 Penalty Records

Search:

Clear all

PENALTIES AND VIOLATION INFORMATION

Lab transactions No:

ID number:
 Student name:
 Email:
 Contact No:

Violations: *
 Penalty Conditions: *
 Violation status: *

Amount to be payed:
 Amount paid:
 Remaining Balance: 0.00

Figure 65: Penalties and Violations

View Students Information
 Students List - View Student Information

Filter by program:

Student List

Enter student's ID No:

Clear all

VIEW STUDENTS INFORMATION

Students name:
 ID number:
 Email:

Contact No:
 Program:
 Department:

Figure 66:View Student Information

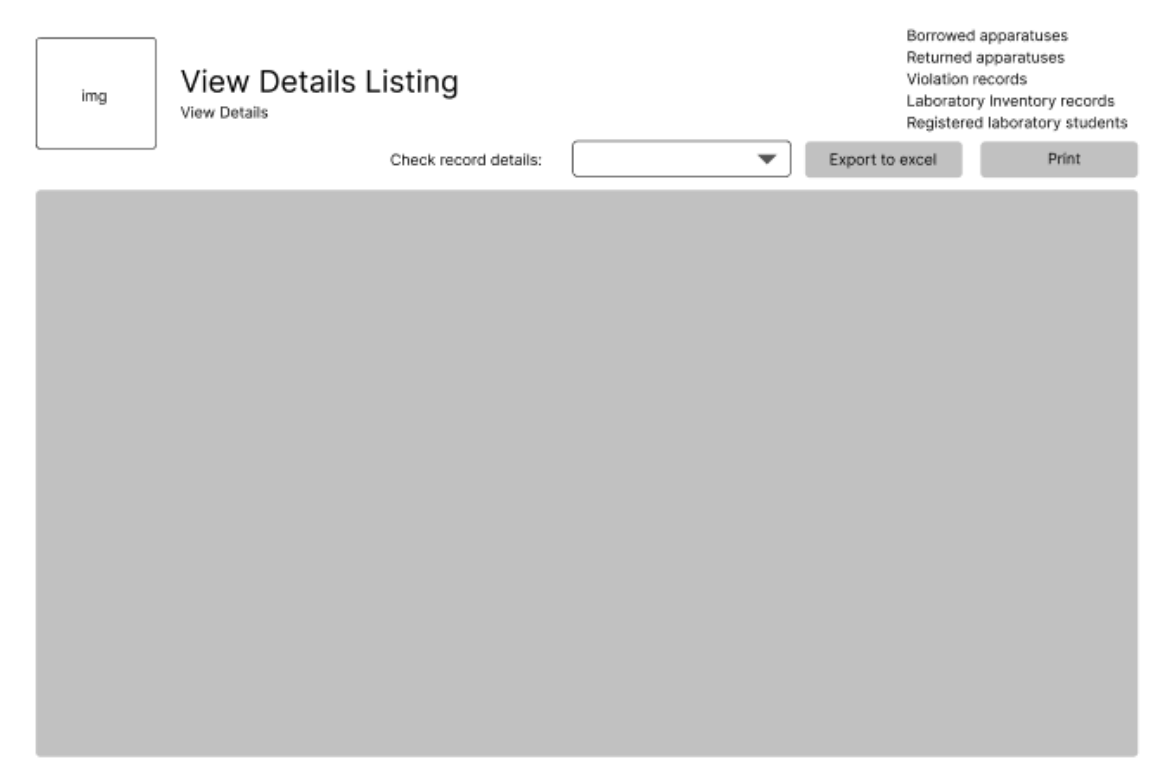


Figure 67:View Details Listing

Appendix D. Sample Program Code

```
using OfficeOpenXml;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.IO;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using static System.Windows.Forms.VisualStyles.VisualStyleElement;

namespace Laboratory_Management_System__Capstone_Project_
{
    public partial class AddApparatus : Form
    {
        private List<string> categoryList = new List<string>();

        private string PlaceholderText = "0.00";
        public AddApparatus()
        {
            InitializeComponent();
            LoadCategories();
            tbPrice.Enter += tbPrice_Enter;
            tbPrice.Leave += tbPrice_Leave;
            // Set the initial placeholder text
            SetPlaceholderText();

            UIHelper.SetRoundedCorners(btnSave, 20);
            UIHelper.SetRoundedCorners(lnklblRefresh, 10);
            UIHelper.SetRoundedCorners(panel2, 30);
            UIHelper.SetRoundedCorners(btnXMark, 30);

            UIHelper.MakeRoundedTextBox(tbAppaName, 10);
            UIHelper.MakeRoundedTextBox(tbModelNum, 10);
            UIHelper.MakeRoundedTextBox(tbPrice, 10);
            UIHelper.MakeRoundedTextBox(tbBrand, 10);
            UIHelper.MakeRoundedTextBox(tbQuantity, 10);
            UIHelper.MakeRoundedTextBox(tbRemarks, 10);

            UIHelper.SetRoundedComboBox(cbStatus, 20);
            UIHelper.SetRoundedComboBox(cbCategory, 20);

            UIHelper.SetRoundedDateTimePicker(dtpDatePurchased, 20);

            UIHelper.SetFormStartLocation(this, 330, 127);

            UIHelper.SetShadow(panel2);
        }
    }
}
```

```

    }

    private void LoadCategories()
    {
        SqlConnection connect = new SqlConnection("data source = LAPTOP-
4KSPM38V; database = LabManagSys;integrated security=True");
        SqlCommand command = new SqlCommand("SELECT CategoryName FROM Category", connect
    );

        connect.Open();
        SqlDataReader reader = command.ExecuteReader();
        while (reader.Read())
        {
            categoryList.Add(reader["CategoryName"].ToString());
        }
        connect.Close();

        cbCategory.Items.Clear();
        cbCategory.Items.AddRange(categoryList.ToArray());
    }

    //Press/Navigate event
    private void SetPlaceholderText()
    {
        if (string.IsNullOrEmpty(tbPrice.Text))
        {
            tbPrice.Text = PlaceholderText;
            tbPrice.ForeColor = Color.Gray;
        }
        else
        {
            tbPrice.ForeColor = Color.Black;
        }
    }

    private void tbPrice_Enter(object sender, EventArgs e)
    {
        if (tbPrice.Text == PlaceholderText)
        {
            tbPrice.Text = "";
            tbPrice.ForeColor = Color.Black;
        }
    }

    private void tbPrice_Leave(object sender, EventArgs e)
    {
        SetPlaceholderText();

        string value = tbPrice.Text;
        bool thereIsDecimal = false;
    }

```

```

        foreach (char c in value)
        {
            if (c == '.')
            {
                thereIsDecimal = true;
            }
        }
        if (thereIsDecimal == false && tbPrice.Text != "")
        {
            tbPrice.Text = value + ".00";
        }
    }
    // Save button event to insert apparatus data, including the selected category
    private void btnSave_Click(object sender, EventArgs e)
    {
        if (tbAppaName.Text != "" && cbStatus.Text != "" && tbQuantity.Text != "" && cbCategory.SelectedIndex != -1 || tbRemarks.Text != "")
        {
            String appa_name = tbAppaName.Text;
            String model_num = tbModelNum.Text == "" ? "N/A" : tbModelNum.Text;
            String date_purch = dtpDatePurchased.Text;
            Decimal price = Decimal.Parse(tbPrice.Text);
            String brand = tbBrand.Text;
            String status = cbStatus.Text;
            Int64 quantity = Int64.Parse(tbQuantity.Text);
            String remarks = tbRemarks.Text;
            String category_name = cbCategory.SelectedItem.ToString();

            SqlConnection connect = new SqlConnection("data source = LAPTOP-4KSPM38V; database = LabManagSys;integrated security=True");
            SqlCommand command = new SqlCommand();
            command.Connection = connect;

            connect.Open();

            // Retrieve CategoryID based on the selected CategoryName
            command.CommandText = "SELECT CategoryID FROM Category WHERE CategoryName = @CategoryName";
            command.Parameters.AddWithValue("@CategoryName", category_name);
            int categoryID = (int)command.ExecuteScalar();

            // Insert apparatus data with the CategoryID
            command.CommandText = "INSERT INTO Inventory ([Apparatus Name], [Model Number], [Date Purchased], [Price], [Brand], [Status], [Quantity], [CategoryID], [Remarks]) " +
                "VALUES (@AppaName, @ModelNum, @DatePurch, @Price, @Brand, @Status, @Quantity, @CategoryID, @Remarks)";
            command.Parameters.Clear();
            command.Parameters.AddWithValue("@AppaName", appa_name);
            command.Parameters.AddWithValue("@ModelNum", model_num);
            command.Parameters.AddWithValue("@DatePurch", date_purch);
            command.Parameters.AddWithValue("@Price", price);
            command.Parameters.AddWithValue("@Brand", brand);
            command.Parameters.AddWithValue("@Status", status);
            command.Parameters.AddWithValue("@Quantity", quantity);

```

```

        command.Parameters.AddWithValue("@CategoryID", categoryID);
        command.Parameters.AddWithValue("@Remarks", remarks);

        command.ExecuteNonQuery();
        connect.Close();

        MessageBox.Show("Apparatus data has been saved and added.", "Success", MessageBoxButtons
        .OK, MessageBoxIcon.Information);

        tbAppaName.Clear();
        tbModelNum.Clear();
        tbPrice.Clear();
        tbBrand.Clear();
        cbStatus.ResetText();
        tbQuantity.Clear();
        dtpDatePurchased.Value = DateTime.Now;
        cbCategory.ResetText();
        tbRemarks.Clear();
    }
    else
    {
        MessageBox.Show("Please fill up the empty fields.", "Error", MessageBoxButtons.OK, Message
        BoxIcon.Warning);
    }
}

private void btnXMark_Click(object sender, EventArgs e)
{
    if (MessageBox.Show("This will remove any UNSAVED data\nDo you still want to cancel this task
?",
        "Confirmation", MessageBoxButtons.YesNo, MessageBoxIcon.Warning) == DialogResult.Yes)
    {
        this.Close();
        //Sets back to 0 to prevent restriction from occuring
        //Dashboard.formRestrict = 0;
    }
}

private void lnklblRefresh_LinkClicked(object sender, LinkLabelLinkClickedEventArgs e)
{
    tbAppaName.Clear();
    tbModelNum.Clear();
    tbPrice.Clear();
    tbBrand.Clear();
    cbStatus.ResetText();
    tbQuantity.Clear();
    dtpDatePurchased.Value = DateTime.Now;
    cbCategory.ResetText();
    tbRemarks.Clear();
}

private void btnMinimize_Click(object sender, EventArgs e)

```

```

        {
            WindowState = FormWindowState.Minimized;
        }
    }
}

```

```

using System.Collections.Generic;
using System.Data.SqlClient;
using System.Windows.Forms;
using System;
using System.Drawing;
using System.Windows.Forms.VisualStyles;
using System.Drawing.Drawing2D;
using System.Linq;

```

```

namespace Laboratory_Management_System__Capstone_Project_
{

```

```

    public partial class Dashboard : Form
    {
        public static int formRestrict;
        private Dictionary<Type, Form> openForms = new Dictionary<Type, Form>();

        public Dashboard()
        {
            InitializeComponent();

            var customRenderer = new UIHelper.CustomToolStripRenderer();
            menuStrip3.Renderer = customRenderer;
            customRenderer.AttachClickHandler(menuStrip3);

            UIHelper.SetShadow(panel1);
            UIHelper.SetRoundedCorners(panel1, 30);
            UIHelper.SetRoundedCorners(panelDashboard, 80);
            UIHelper.SetRoundedCorners(panelRules, 30);
            UIHelper.SetRoundedCorners(pictureBox1, 40);
            UIHelper.SetRoundedCorners(btnInventoryShortcut, 30);
            UIHelper.SetRoundedCorners(btnStudentListShortcut, 30);

            UIHelper.SetRoundedCorners(ShowCountPanel, 20);

            UIHelper.SetGradientBackground(panel1, Color.FromArgb(5, 21, 101), Color.FromArgb(20, 57, 175), LinearGradientMode.Horizontal);
            UIHelper.SetGradientBackground(panelDashboard, Color.FromArgb(5, 21, 101), Color.FromArgb(20, 57, 175), LinearGradientMode.Horizontal);
        }

        private void Dashboard_Load(object sender, EventArgs e)
        {
            int currentAccountId = GetCurrentAccountId();

```

```

        if (currentAccountId == -1)
        {
            MessageBox.Show("Invalid session. Please log in again.", "Session Error", MessageBoxButtons.
OK, MessageBoxIcon.Error);
            Form1 login = new Form1();
            login.Show();
            this.Hide();
            return;
        }

        string userRole = GetUserRole(currentAccountId);

        lblIDNumber.Text = $"{Form1.Session.IDNumber} - " + $"{Form1.Session.Role}";
        lblFirstName.Text = $"{Form1.Session.FirstName}";

        // Restrict access based on user role
        if (userRole == "Personnel")
        {
            penaltyRecordsToolStripMenuItem.Visible = false; // Hide access to PenaltiesRecords for Person
nel
            updateTransactionsToolStripMenuItem.Visible = false; // Hide access to UpdateTransactions for
Personnel
            btnAccounts.Visible = false; // Hide access to Accounts for Personnel
        }

        ShowShortcutButtons();
        ShowApparatusCount();
    }

    private void picBoxBC_Click(object sender, EventArgs e)
    {
        // Prompt the user for confirmation
        if (MessageBox.Show("Are you sure you want to close all forms?", "Confirm Close", MessageBoxButtons.YesNo) == DialogResult.Yes)
        {
            // Remove any forms that are currently being shown in the panelContainer
            foreach (Control control in panelContainer.Controls)
            {
                if (control is Form)
                {
                    control.Dispose();
                }
            }

            // Show the Dashboard controls
            ShowShortcutButtons();
            ShowApparatusCount();

            // Refresh the UI to ensure everything is updated
            this.Refresh();
        }
    }

    private int GetCurrentAccountId()

```

```

    {
        return Form1.Session.AccountID; // Retrieve the stored user ID from the session
    }

    private string GetUserRole(int accountId)
    {
        string role = string.Empty;

        string connectionString = "data source = LAPTOP-
4KSPM38V; database = LabManagSys; integrated security=True";
        string query = @"
        SELECT r.RoleName
        FROM Accounts a
        INNER JOIN Roles r ON a.RoleID = r.RoleID
        WHERE a.AccountID = @AccountID";

        using (SqlConnection connection = new SqlConnection(connectionString))
        {
            SqlCommand command = new SqlCommand(query, connection);
            command.Parameters.AddWithValue("@AccountID", accountId);

            try
            {
                connection.Open();
                object result = command.ExecuteScalar();
                if (result != null)
                {
                    role = result.ToString();
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show($"An error occurred: {ex.Message}", "Error", MessageBoxButtons.OK, M
essageBoxIcon.Error);
            }
        }

        return role;
    }

    // Method to display forms in the panel container
    public void ShowFormInPanel(Form form)
    {
        // Remove any forms that are currently being shown in the panelContainer
        foreach (Control control in panelContainer.Controls)
        {
            if (control is Form)
            {
                control.Dispose();
            }
        }

        // Hide the shortcut buttons

```

```

        btnInventoryShortcut.Visible = false;
        btnStudentListShortcut.Visible = false;
        lblShortcut.Visible = false;
        lblFirstName.Visible = false;
        lblIDNumber.Visible = false;

        // Hide the labels
        apparatusCountLabel.Visible = false;
        studentCountLabel.Visible = false;
        borrowedApparatusCountLabel.Visible = false;
        returnedApparatusCountLabel.Visible = false;
        lblOverview.Visible = false;
        ShowCountPanel.Visible = false;
        lblTitle.Visible = false;
        panelDashboard.Visible = false;
        panelRules.Visible = false;

        form.TopLevel = false;
        form.FormBorderStyle = FormBorderStyle.None;
        form.Dock = DockStyle.Fill;
        panelContainer.Controls.Add(form);
        form.Show();
    }

    /*public void ShowFormInPanel(Form form)
    {
        panelContainer.Controls.Clear();
        form.TopLevel = false;
        form.FormBorderStyle = FormBorderStyle.None;
        form.Dock = DockStyle.Fill;
        panelContainer.Controls.Add(form);
        form.Show();
    }*/

    // Method to handle the Form closed event
    private void Form_FormClosed(object sender, FormClosedEventArgs e)
    {
        Form form = sender as Form;
        if (form != null)
        {
            openForms.Remove(form.GetType());
            form.FormClosed -= Form_FormClosed; // Unsubscribe from the event
            form.Dispose(); // Release resources
        }
    }

    // Method to restrict multiple instances of a form
    private void ShowRestrictedForm(Form form)
    {
        if (openForms.ContainsKey(form.GetType()))
        {
            MessageBox.Show($"{form.Text} is already open.", "Information", MessageBoxButtons.OK, MessageBoxIcon.Information);
            return;
        }
    }

```

```

    }

    form.FormClosed += Form_FormClosed;
    form.Show();
    openForms[form.GetType()] = form;
}

private void dashboardDefaultToolStripMenuItem_Click(object sender, EventArgs e)
{
    ShowShortcutButtons();
    ShowApparatusCount();
}
private void addANewApparatusToolStripMenuItem_Click(object sender, EventArgs e)
{
    AddApparatus addApparatus = new AddApparatus();
    ShowRestrictedForm(addApparatus);
}

private void viewApparatusListToolStripMenuItem_Click(object sender, EventArgs e)
{
    InventoryList viewApp = new InventoryList();
    ShowFormInPanel(viewApp);
}

private void addAStudentToolStripMenuItem_Click(object sender, EventArgs e)
{
    AddStudent addStudent = new AddStudent();
    ShowRestrictedForm(addStudent);
}

private void viewStudentsInformationToolStripMenuItem_Click(object sender, EventArgs e)
{
    ViewStudentInformation viewStudents = new ViewStudentInformation();
    ShowFormInPanel(viewStudents);
}

private void borrowTransactionToolStripMenuItem_Click(object sender, EventArgs e)
{
    BorrowingApparatus borrowApparatus = new BorrowingApparatus();
    ShowRestrictedForm(borrowApparatus);
}

private void returnRecordsToolStripMenuItem_Click(object sender, EventArgs e)
{
    ReturnApparatus returnApparatus = new ReturnApparatus();
    ShowFormInPanel(returnApparatus);
}

private void comToolStripMenuItem_Click(object sender, EventArgs e)
{
    TransactionDetails details = new TransactionDetails();
    ShowFormInPanel(details);
}

```

```

private void penaltyRecordsToolStripMenuItem_Click(object sender, EventArgs e)
{
    PenaltiesRecords penalty = new PenaltiesRecords();
    ShowFormInPanel(penalty);
}

private void updateTransactionsToolStripMenuItem_Click(object sender, EventArgs e)
{
    VerificationForUpdate verification = new VerificationForUpdate(this);
    ShowRestrictedForm(verification);
}

private void lnklblLogOut_LinkClicked(object sender, LinkLabelLinkClickedEventArgs e)
{
    if (MessageBox.Show("Are you sure you want to log out?", "Confirmation",
        MessageBoxButtons.YesNo, MessageBoxIcon.Warning) == DialogResult.Yes)
    {
        Form1 login = new Form1();
        login.Show();
        this.Hide();
    }
}

private void btnInventoryShortcut_Click(object sender, EventArgs e)
{
    InventoryList viewApp = new InventoryList();
    ShowFormInPanel(viewApp);
}

private void btnStudentListShortcut_Click(object sender, EventArgs e)
{
    ViewStudentInformation viewStudents = new ViewStudentInformation();
    ShowFormInPanel(viewStudents);
}

private void ShowShortcutButtons()
{
    btnInventoryShortcut.Visible = true;
    btnStudentListShortcut.Visible = true;
    lblShortcut.Visible = true;

    apparatusCountLabel.Visible = true;
    studentCountLabel.Visible = true;
    borrowedApparatusCountLabel.Visible = true;
    returnedApparatusCountLabel.Visible = true;
    lblOverview.Visible = true;
    ShowCountPanel.Visible = true;
    lblTitle.Visible = true;
    panelRules.Visible = true;
    lblIDNumber.Visible = true;
    lblFirstName.Visible = true;
    panelDashboard.Visible = true;
}

```

```

private void ShowApparatusCount()
{
    // Update the apparatus count label
    UpdateApparatusCountLabel(apparatusCountLabel);

    // Update the student count label
    UpdateStudentCountLabel(studentCountLabel);

    // Update the borrowed apparatus count label
    UpdateBorrowedApparatusCountLabel(borrowedApparatusCountLabel);

    // Update the returned apparatus count label
    UpdateReturnedApparatusCountLabel(returnedApparatusCountLabel);

    ShowCountPanel.Visible = true;
}

private void UpdateApparatusCountLabel(Label apparatusCountLabel)
{
    // Connect to the database and retrieve the apparatus count
    string connectionString = "data source = LAPTOP-
4KSPM38V; database = LabManagSys; integrated security=True";
    string query = "SELECT COUNT(*) FROM Inventory";

    using (SqlConnection connection = new SqlConnection(connectionString))
    {
        connection.Open();
        SqlCommand command = new SqlCommand(query, connection);
        int apparatusCount = (int)command.ExecuteScalar();

        // Update the apparatus count label
        apparatusCountLabel.Text = $"{apparatusCount}";
    }
}

private void UpdateStudentCountLabel(Label studentCountLabel)
{
    // Connect to the database and retrieve the student count
    string connectionString = "data source = LAPTOP-
4KSPM38V; database = LabManagSys; integrated security=True";
    string query = "SELECT COUNT(*) FROM Students";

    using (SqlConnection connection = new SqlConnection(connectionString))
    {
        connection.Open();
        SqlCommand command = new SqlCommand(query, connection);
        int studentCount = (int)command.ExecuteScalar();

        // Update the student count label
        studentCountLabel.Text = $"{studentCount}";
    }
}

```

```

    }

    private void UpdateBorrowedApparatusCountLabel(Label borrowedApparatusCountLabel)
    {
        // Connect to the database and retrieve the borrowed apparatus count
        string connectionString = "data source = LAPTOP-
4KSPM38V; database = LabManagSys; integrated security=True";
        string query = "SELECT COUNT(*) FROM BorrowReturnTransaction WHERE Date_Returned IS
NULL";

        using (SqlConnection connection = new SqlConnection(connectionString))
        {
            connection.Open();
            SqlCommand command = new SqlCommand(query, connection);
            int borrowedApparatusCount = (int)command.ExecuteScalar();

            // Update the borrowed apparatus count label
            borrowedApparatusCountLabel.Text = $"{borrowedApparatusCount}";
        }
    }

    private void UpdateReturnedApparatusCountLabel(Label returnedApparatusCountLabel)
    {
        // Connect to the database and retrieve the returned apparatus count
        string connectionString = "data source = LAPTOP-
4KSPM38V; database = LabManagSys; integrated security=True";
        string query = "SELECT COUNT(*) FROM BorrowReturnTransaction WHERE Date_Returned IS
NOT NULL";

        using (SqlConnection connection = new SqlConnection(connectionString))
        {
            connection.Open();
            SqlCommand command = new SqlCommand(query, connection);
            int returnedApparatusCount = (int)command.ExecuteScalar();

            // Update the returned apparatus count label
            returnedApparatusCountLabel.Text = $"{returnedApparatusCount}";
        }
    }

    public void NavigateBackToDashboard()
    {
        // Remove the current form from the panelContainer
        panelContainer.Controls.Clear();

        // Show the shortcut buttons
        btnInventoryShortcut.Visible = true;
        btnStudentListShortcut.Visible = true;
        lblShortcut.Visible = true;

        // Show the labels
        apparatusCountLabel.Visible = true;
        studentCountLabel.Visible = true;
    }

```

```

        borrowedApparatusCountLabel.Visible = true;
        returnedApparatusCountLabel.Visible = true;
        lblOverview.Visible = true;
        ShowCountPanel.Visible = true;
        lblTitle.Visible = true;
    }

    private void dashboardtoolStripMenu_Click(object sender, EventArgs e)
    { // Remove any forms that are currently being shown in the panelContainer
        foreach (Control control in panelContainer.Controls)
        {
            if (control is Form)
            {
                control.Dispose();
            }
        }

        // Show the Dashboard controls
        ShowShortcutButtons();
        ShowApparatusCount();

        // Refresh the UI to ensure everything is updated
        this.Refresh();
    }

    private void btnAccounts_Click(object sender, EventArgs e)
    {
        Accounts account = new Accounts();
        account.Show();
    }
}

```

Appendix E. System Usability Scale

USER ACCEPTANCE FORM

SYSTEM USABILITY SCALE

TITLE OF THE CAPSTONE PROJECT

DATE

NAME OF THE CLIENT/USER

SIGNATURE

Instruction: Please check the box that reflects your response to each statement.

	Strongly Disagree				Strongly Agree
	1	2	3	4	5
1. I think that I would like to use this product frequently.	☐	☐	☐	☐	☐
2. I found the product unnecessarily complex.	☐	☐	☐	☐	☐
3. I thought the product was easy to use.	☐	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this product.	☐	☐	☐	☐	☐
5. I found the various functions in the product were well integrated.	☐	☐	☐	☐	☐
6. I thought there was too much inconsistency in this product.	☐	☐	☐	☐	☐
7. I imagine that most people would learn to use this product very quickly.	☐	☐	☐	☐	☐
8. I found the product very awkward to use.	☐	☐	☐	☐	☐
9. I felt very confident using the product.	☐	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this product.	☐	☐	☐	☐	☐

Comments/Suggestions

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Appendix F. Plagiarism Result



TURNITIN RECEIPT

MEMBERS: Tricea Mae O. Cortes
Raul Vincent Rodriguez
Gladys Mae Caballero
Danica A. Basalo

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Curriculum Vitae

PERSONAL DATA

Name : Tricea Mae Cortes
Address : V Rama Cebu, City.
Date and Place of Birth : March 19, 2001 | Cebu City
Civil Status : Single
Religion : Bible Baptist
Mother's name : Raquel Cortes
Father's name : Jimmy Cortes



EDUCATIONAL BACKGROUND

Tertiary Level

Name of School : Asian College of Technology
Degree : Bachelor of Science in Information Technology
Address : P.del Rosario and Leon Kilat Sts., Cebu City

Secondary Level

Name of School : Dalaguete National Highschool
Address : Poblacion, Dalaguete, Cebu
Year Graduated : 2018

Elementary Level

Name of School : Dalaguete Central Elementary School
Address : Poblacion, Dalaguete, Cebu
Year Graduated : 2014

PERSONAL DATA

Name : Raul Vincent Rodriguez
Address : Tabunok, Talisay City, Cebu
Date of Birth : February 2, 2002
Civil Status : Single
Religion : Roman Catholic
Father's Name : Federico R. Juguilon
Mother's Name : Michelle R. Juguilon



EDUCATIONAL BACKGROUND

Tertiary Level

Name of School : Asian College of Technology
Degree : Bachelor of Science in Information Technology
Address : Corner P Del Rosario & Leon Kilat St.

Secondary Level

Name of School : Divino Amore Academy
Address : Talisay City, Cebu
Year Graduated : 2019

Elementary Level

Name of School : Exceed Learning Center
Address : Naga City, Cebu
Year Graduated : 2015

PERSONAL DATA

Name : Danica A. Basalo
Address : Calajo-an, Minglanilla,
Cebu City, 6046
Date of Birth : November 17, 2002
Civil Status : Single
Religion : Roman Catholic
Father's Name : Danilo Basalo
Mother's Name : Jocel Basalo



EDUCATIONAL BACKGROUND

Tertiary Level

Name of School : Asian College of Technology
Degree : Bachelor of Science in Information Technology
Address : Corner P Del Rosario & Leon Kilat St.

Secondary Level

Name of School : University of the Visayas
Address : Tiber, Minglanilla, Cebu
Year Graduated : 2019

Elementary Level

Name of School : Calajo-An Elementary School
Address : Lower Calajo-an, Minglanilla, Cebu
Year Graduated : 2014

PERSONAL DATA

Name : Gladys Mae Caballero
Address : Tubod, Badian, Cebu
Date and Place of Birth : November 27, 2002
Civil Status : Single
Religion : Roman Catholic
Mother's name : Leonora E. Caballero
Father's name : Celestino C. Caballero



EDUCATIONAL BACKGROUND

Tertiary Level

Name of School : Asian College of Technology
Degree : Bachelor of Science in Information Technology
Address : P.del Rosario and Leon Kilat Sts., Cebu City

Secondary Level

Name of School : Tubod, National High School
Address : Tubod, Badian, Cebu
Year Graduated : 2019

Elementary Level

Name of School : Tubod Elementary School
Address : Tubod, Badian, Cebu
Year Graduated : 2015